Notes on Harmonic Analysis on Finite Abelian Groups

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1 Introduction

These notes are an introduction to harmonic analysis for my Directed Reading Program (DRP) and REU groups. They expect only a background in linear algebra, though familiarity with group theory will help digest the speedy introduction in the first three sections. There is also quite a bit of category theory lurking in this topic; I have acknowledged it where appropriate, but no background in it is expected. The goal of these notes is to introduce the ideas necessary for reading the papers [AK62] and [GM95], as well as to provide a companion to those papers in the later sections.

There are two important caveats. First, these notes are aimed at preparing the reader for research in higher-order autocorrelations, but they are *not* intended to contain the minimal content necessary to begin this research. I have an ulterior motive: to teach you something about the theory of harmonic analysis that will benefit you in all of your future endeavors too. The second "obstacle" is the presence of many exercises, some of which contain crucial ideas. Once again, this comes from a (not-so-)hidden agenda: to impress upon you that reading a text is an interactive experience. As such, the exercises should be regarded as mandatory, except when marked otherwise. A benefit of frequent meetings is that you will not stay stuck on one exercise for too long.

Our primary concern will be the Fourier transform. The Fourier transform on a finite abelian group G arises as a clever change of basis on the space of functions on G, into a basis of so-called *characters* which have a distinguished algebraic role. It so happens that these characters are (more or less) the complex exponentials $e_\gamma(x) := e^{2\pi i\gamma x}$, which wear many different hats. For instance, this explains the usefulness of Fourier analysis in differential equations. If we wish to relax the condition that G be finite, the space of functions on G becomes infinite-dimensional, and so the requirements that must now be placed on G are *topological* so that sequences of functions converge as desired. Thus, in the case of G finite, the emphasis is largely algebraic; in the case of G infinite, the algebraic picture is essentially unchanged, but the analytic picture is much more

complicated. Our focus will be on fleshing out the algebraic picture (which also effectively describes the end result of the analytic undertaking in the more general setting) and hinting towards the difficulties and tools one might encounter in the more general setting.

2 Groups

Groups are the prototypical set-with-structure that form the backbone of modern algebra. Frequently, across arithmetic, geometry, and all sorts of more specialized mathematical disciplines, one has a set of objects G with some way to combine them (adding numbers, composing functions, etc.). Groups tread the fine line between being general enough to be interesting but structured enough to be tractable.

Definition 2.1. A group is a set G with a binary operation $\cdot : G \times G \to G$ satisfying the following axioms:

(G1) There is an *identity element* $1 \in G$ such that $1 \cdot g = g \cdot 1 = g$ for all $g \in G$. When we want to emphasize the group G (e.g. when we are considering multiple groups at the same time), we might write 1_G.

(G2) Every element $g \in G$ has an *inverse element* g^{-1} such that $g \cdot g^{-1} = g^{-1} \cdot g = 1$.

(G3) Multiplication is associative: given $a, b, c \in G$, we have $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.

The images you should have in mind are of the typical arithmetic situations: the nonzero real numbers with binary operation given by multiplication (typically written $\mathbb{R}^\times$), as well as the integers $\mathbb{Z}$ with addition as the operation. (Why is multiplication not a valid group operation on $\mathbb{Z}$? Which axiom is violated?)

There is one axiom conspicuously missing from our definition which is nevertheless satisfied by both of our examples. We do **not** demand that a group operation is commutative, i.e. that $a \cdot b = b \cdot a$. A group with a commutative group operation is called **abelian**. Inspired by the examples of $\mathbb{Z}$ and $\mathbb{R}$ under the operation of addition, we usually write abelian groups additively. For convenience, let us reformulate the group axioms in this way.

Definition 2.2. An **abelian group** is a set G with a binary operation $+ : G \times G \to G$ satisfying the following axioms:

(G1) There is an identity element $0 \in G$ such that $0 + g = g + 0 = g$ for all $g \in G$.

(G2) Every element $g \in G$ has an inverse element $-g$ such that $g + (-g) = (-g) + g = 0$.

(G3) Addition is associative: given $a, b, c \in G$, we have $a + (b + c) = (a + b) + c$.

(G4) Addition is commutative: given $a, b \in G$, we have $a + b = b + a$.

Let us give one more reformulation of the notion of a group, which is preferable to enjoyers of abstract nonsense. It will be useful to us later. If your head is already spinning, feel free to skip to the examples below, but be prepared to return to this definition when we need it.

Definition 2.3. A group G is a set with three operations:

(i) $1 : \{\bullet\} \to G$, where we write $1 = 1(\bullet) \in G$. (This operator "picks out the identity" in G.)

(ii) $i : G \to G$, where we write g^{-1} for $i(g)$. (This operator specifies the inverse of each element.)

(iii) $m : G \times G \to G$, where we write $g \cdot h$ for $m(g, h)$. (This is the multiplication operator.)

Exercise 2.4 (discouraged for a first read but great for a second one)**.** Reformulate the group axioms (G1)-(G3) in terms of the operations above. To get you started: (G1) says that multiplying by the identity element gives the identity function, i.e. that $m \circ (\mathrm{id}_G \times 1) = \mathrm{id}_G$. This is the same as saying that the following diagram **commutes**, meaning that following either path from the top-left to the bottom-right gives the same result:

$$\begin{tikzcd}
G \times \{\bullet\} \arrow[r, "(g,\bullet)\mapsto g"] \arrow[d, "\mathrm{id}\times 1"'] & G \arrow[d, "\mathrm{id}"] \\
G \times G \arrow[r, "m"'] & G
\end{tikzcd}$$

This presentation of a group aligns with the process (typical of category theory) of presenting an entity via (a) its data, (b) with some structure, (c) subject to some coherence conditions. If you are inclined towards category theory, you will appreciate that this allows us to define a **group object** in any category as an object in that category with a collection of morphisms as above.

Let us descend from our 30,000-foot abstraction altitude and look at some concrete examples.

Example 2.5 (the integers)**.** The set of integers $\mathbb{Z}$ with the operation of addition forms an abelian group, with identity 0. The inverse of n is $-n$. This makes the notation of our definition of abelian groups much clearer.

Example 2.6 (the integers $\mod n$)**.** $\mathbb{Z}_n$ is the group of integers $\mod n$, i.e. where two integers are considered the same if they differ by a multiple of n. Some authors write $\mathbb{Z}/n\mathbb{Z}$ for this group, with good reason (see Example 3.18), but it is more economical for us to write $\mathbb{Z}_n$.

Example 2.7 (the real line and complex plane)**.** The set of real numbers $\mathbb{R}$ forms a group under addition. Alternatively, the set of nonzero real numbers under multiplication forms a group, denoted $\mathbb{R}^\times$. Similarly, the nonzero complex numbers form a group under multiplication, called $\mathbb{C}^\times$.

Example 2.8 (the circle group)**.** Let $S^1 = \{z \in \mathbb{C} : |z| = 1\}$ be the unit circle inside the complex plane $\mathbb{C}^\times$, with operation given by multiplication of complex numbers. Since complex numbers have a polar form, the condition $|z| = 1$ guarantees $z = e^{i\theta}$ for some θ, and multiplication of two elements of S^1 corresponds to rotating one by the angle of the other.

Then 2π-periodicity of the complex exponential suggests an alternative definition: S^1 is $\mathbb{R}/\mathbb{Z}$, the set of real numbers which are considered equivalent if they differ by an integer. In other words, $\mathbb{R}/\mathbb{Z}$ is identified with the interval $[0, 1)$, where addition is carried out $\mod 1$. (This is, after scaling, the same as the interval $[0, 2\pi)$ with addition $\mod 2\pi$, making the analogy with the unit circle clearer. You can picture an interval "with the endpoints tied together.")

When we have a notion of two groups being "the same," we will see that these two presentations of S^1 are the same; see Exercise 4.4.

Example 2.9. The first non-abelian group we will see is the **symmetric group** S_n. For any n, S_n is the set of bijections from $\{1, ..., n\}$ to itself, with the operation of composition. This group is important and also complicated, two statements justified by Cayley's Theorem 4.6.

Exercise 2.10. Check the group axioms for as many of the above examples as you can before developing a headache.

We finish this section with an important definition that will find utility when we discuss group homomorphisms.

Definition 2.11. Let $H \subset G$ be a nonempty subset of a group G. We say that H is a **subgroup** if H is a group with respect to the operation it inherits from G. Explicitly, H is a subgroup if:

- The identity 1 of G is in H;
- If $h \in H$, then $h^{-1} \in H$;
- If $g, h \in H$, then $gh \in H$.

Here the identity, inverses, and multiplication are all the ones from G, applied to elements in $H \subset G$.

Exercise 2.12. Check that $H \subset G$ is a subgroup if and only if it is nonempty and, whenever $g, h \in H$, we also have $gh^{-1} \in H$.

3 Homomorphisms

There is a pervasive philosophy throughout mathematics that an object should be understood in terms of the functions ("maps" or "morphisms") into and out of it. Whenever you encounter a new type of object, you should consider what is the "right" notion of a map between such objects. In this section, we will study maps between groups, which are an archetypal example: they are maps of sets which "preserve the group structure" in the following way.

Definition 3.1. Let G and H be groups. A function $f : G \to H$ is called a **group homomorphism** ("hom" for short) if $f(xy) = f(x)f(y)$ for all $x, y \in G$. Note particularly that the product xy on the left uses the multiplication in G, while the product on the right uses the product in H.

Exercise 3.2. Show that, if f is a group homomorphism, $f(1_G) = 1_H$ and $f(x^{-1}) = f(x)^{-1}$. (Hint: For the former, consider $f(1_G 1_G)$.)

The utility of such an idea can really only be seen by immersion, but one hopes that by incorporating some knowledge of the group operation into our functions, we gain a substantial amount of control over the behavior of these functions. Functions between sets are wild, in that they can send any element to any other element with no regard for what those elements "mean." We imagine that any "reasonable" or "natural" map between groups already takes into account the group structure, so will be a group homomorphism already. At this point, these are only hopes, but they will more or less turn out to be true.

Let us now look at a few classes of particularly important examples.

Example 3.3. Let $H \subset G$ be a subgroup. There is a map $\iota : H \to G$ which sends $h \in H$ to h viewed now as an element of G. This is called the **inclusion** of H into G.

Exercise 3.4. Check that the inclusion of a subgroup is an injective group homomorphism.

The inclusion of a subgroup is the archetypal injective group homomorphism. We will see later (Theorem 4.3) that, in some sense, every injective group homomorphism is the inclusion of some subgroup. Let us turn our attention briefly to an important class of subgroups, while we are on the subject, which we will be heavily interested in later.

Example 3.5. The most important class of subgroups is given by the **kernel** of a group homomorphism,

$$\ker f = \{g \in G : f(g) = 1\}.$$

That is, the kernel of a map is "everything it kills." (As a possibly familiar example: viewing a vector space as a group under addition, the kernel of a linear map is the kernel in this sense too.)

Exercise 3.6 (absolutely mandatory)**.** Show that a group homomorphism $f : G \to H$ is injective if and only if $\ker f = \{1_G\}$. Practically, this provides a way to check for injectivity much more easily. Philosophically, it exemplifies the idea that "global behavior is captured by local behavior at the identity." (This is a hint towards the solution: take a failure of injectivity $f(g) = f(h)$, and translate it to a failure of injectivity at the identity, $f(gh^{-1}) = 1$.)

Definition 3.7. A subgroup $H \subset G$ is **normal** if, for every $h \in H$ and $g \in G$, the **conjugate** ghg^{-1} is also in H. Note that, in an abelian group, every subgroup is normal, as $ghg^{-1} = gg^{-1}h = h$.

Exercise 3.8. Show that the kernel of a group homomorphism is always normal.

The situation for surjective group homomorphisms is much more complicated. Before we consider them, let us look at *bijective* group homomorphisms, i.e. ones which are both injective and surjective.

Definition 3.9. A bijective group homomorphism is called an **isomorphism**. If $f : G \to H$ is an isomorphism, then we call G and H **isomorphic** and write $G \cong H$.

Remark 3.10. You should imagine isomorphic groups as being "the same," with elements simply relabeled. Indeed, the bijection gives the "relabeling," i.e. tells you to think of g and $f(g)$ as the same. That $f(gh) = f(g)f(h)$ tells you that, if you relabel every element by its image under f, you also relabel products (in G) to the correct products in H. That is, the multiplication in H is the same as multiplication in G, after this relabeling.

Now, let us look at some examples of surjective group homomorphisms. We will get a sense of how they behave, which will lead us to a particularly important class of surjective homomorphisms, which again by Theorem 4.3 will turn out to be all of them.

Example 3.11. There is a surjective group homomorphism $f : \mathbb{Z} \to \mathbb{Z}_n$ given by "reduction mod n," i.e. $k \mapsto k \mod n$. If you like, check that this is indeed a homomorphism. It is certainly surjective, but we can understand the preimages $f^{-1}(m)$ in more detail: m is sent to $m' \in \mathbb{Z}_n$ if and only if $m \equiv m' \mod n$, i.e. iff $m - m'$ is a multiple of n. In particular, the elements of $\mathbb{Z}$ sent to $0 \in \mathbb{Z}_n$ are the multiples of n, denoted $n\mathbb{Z}$. Furthermore, if $m \in f^{-1}(m')$, i.e. $f(m) = m'$, then $m = m' + kn$ for some $k \in \mathbb{Z}$. Hence all the preimages $f^{-1}(m')$ are in bijection with $\ker f = \{nz : z \in \mathbb{Z}\} = n\mathbb{Z}$ via $m' + kn \mapsto kn \in n\mathbb{Z}$.

Example 3.12. There is a surjective group homomorphism $f : \mathbb{R} \to S^1$ given by $\theta \mapsto e^{2\pi i\theta}$. Once again, check that this is a group homomorphism if you are so inclined. We can play the same game with kernels as above: if $\theta \in \ker f$, then $e^{2\pi i\theta} = 1$, so θ is an integer. For every $y \in S^1$, there is an angle $\phi \in [0, 1)$ such that $e^{2\pi i\phi} = y$, and so elements x with $f(x) = y$ are in bijection with $\ker f$ by adding or subtracting an integer part from the "fundamental solution" ϕ.

We will now describe the construction that produces the examples above, and many more besides.

Definition 3.13. Let $H \subset G$ be a subgroup. Define an equivalence relation $\sim$ on G by saying $g_1 \sim g_2$ if $g_1^{-1}g_2 \in H$. Let $gH = [g]_\sim = \{g' \in G : g' \sim g\}$ be the equivalence class of g under this equivalence relation. We call gH the **coset** of g mod H.

Exercise 3.14. Show that $\sim$ (as described above) is an equivalence relation and that the equivalence classes are given by $gH = \{gh : h \in H\}$.

Exercise 3.15. Suppose that S is a set with an equivalence relation $\sim$. Show that S is partitioned into disjoint equivalence classes, i.e. that for equivalence classes $[x]_\sim = \{s \in S : x \sim s\}$, the equivalence classes $[x]_\sim$ and $[y]_\sim$ are either disjoint or equal as subsets of S. Deduce that $x \sim y$ if and only if $[x]_\sim = [y]_\sim$.

Definition 3.16 (continuation)**.** Let G/H (read "G mod H") be the set of equivalence classes under the relation $\sim$ defined above, i.e. $G/H = \{gH : g \in G\}$. G/H is called the **quotient** of G by H.

So far, our definition of the quotient works perfectly fine for any set with an equivalence relation: we have simply broken G up into its equivalence classes under some relation. (Indeed, later on, we will take quotients of sets which are not groups.) We would now like to define a natural group operation on G/H which it "inherits" from G. It turns out that this is only possible when H is a normal subgroup of G.

Proposition 3.17. *Suppose H is a normal subgroup of G, and let G/H be the quotient. For two cosets g_1H and g_2H in G/H, define their product by*

$$g_1H \cdot g_2H = g_1g_2H.$$

This operation makes G/H into a group.

Proof. At issue is **well-definedness**: an equivalence class gH has lots of representatives, i.e. $gH = g'H$ whenever $g' \in gH$, or equivalently whenever $g \sim g'$. We need to check that our definition of the product does not depend on a choice of representatives, i.e. that if $g_1H = g_1'H$, then $g_1H \cdot g_2H = g_1'H \cdot g_2H$. Since an element of the equivalence class g_1H has the form g_1h for some $h \in H$, it suffices to check that

$$g_1hH \cdot g_2H = g_1H \cdot g_2H \tag{1}$$

and

$$g_1H \cdot g_2hH = g_1H \cdot g_2H \tag{2}$$

whenever $h \in H$. (That is, we can change the representative for either term in the product.)

Checking (2) is easier: by definition,

$$g_1H \cdot g_2hH = (g_1g_2h)H,$$

but $g_1g_2hH = g_1g_2H$ as $h \in H$ implies $g_1g_2 \sim g_1g_2h$ (i.e. they differ by an element of H). This tells us that

$$g_1H \cdot g_2hH = g_1H \cdot g_2H = g_1g_2H$$

as desired.

Checking (1) is harder, and requires normality: we want

$$g_1hg_2H = g_1g_2H,$$

or equivalently $g_1hg_2 \sim g_1g_2$. By the definition of the equivalence relation, this means

$$(g_1hg_2)^{-1}g_1g_2 \in H,$$

and since $(g_1hg_2)^{-1} = g_2^{-1}h^{-1}g_1^{-1}$, it is the same to check

$$(g_1hg_2)^{-1}g_1g_2 = g_2^{-1}h^{-1}g_2 \in H.$$

But $h^{-1} \in H$ if and only if $h \in H$, and since this is meant to hold for all $h \in H$ and all $g_2 \in G$, this is precisely the condition of normality! Hence the operation is well-defined if and only if H is normal in G.

□

This is rather abstract and frequently confusing to students on their first brush with group theory. It may be illuminating to revisit some of our earlier examples.

Example 3.18. Consider the subgroup $n\mathbb{Z} = \{nz : z \in \mathbb{Z}\}$ in $\mathbb{Z}$. Since $\mathbb{Z}$ is abelian, every subgroup is normal. The equivalence relation for the quotient (see Definition 3.13) says that $x \sim y$ if and only if $y - x \in n\mathbb{Z}$, i.e. if and only if $y - x$ is a multiple of n (equivalently, if $x - y$ is; this is the more common formulation). Hence $\mathbb{Z}/n\mathbb{Z}$ is the group $\mathbb{Z}_n$ of integers mod n, its operation inherited from the quotient is precisely the operation in $\mathbb{Z}_n$, and the map f described in Example 3.11 sends an element of $\mathbb{Z}$ to its equivalence class in the quotient.

This latter idea is so important that we give it its own name.

Definition 3.19. Let $H \subset G$ be a subgroup. The **natural projection onto the quotient** is the map $\pi : G \to G/H$ with $\pi(g) = gH$.

Remark 3.20. The natural projection π is a group homomorphism if and only if the natural operation on G/H is well-defined, i.e. if and only if H is normal in G. In this case, observe that $\ker \pi = H$, i.e. the elements sent to the identity class $1H \in G/H$ are exactly the elements of H (which differ from 1 by an element of H, namely themselves). This explains the kernel computation in Example 3.11.

The remaining part of Example 3.11 yet to be explained is that preimages $\pi^{-1}(k)$, i.e. cosets $k + n\mathbb{Z}$, are in bijection with $\ker f$ by $k + nz \mapsto nz$. This is the content of Lagrange's theorem 4.1, but you can think about it now.

You are also encouraged to think through Example 3.12 in this new light, but it will be easier later, and will be formulated as Exercise 4.4 then.

Remark 3.21. It is often useful to think of the equivalence relation defining the quotient as $g \sim h \iff \pi(g) = \pi(h)$ for π the natural projection. We will use this perspective to generalize the quotient process later: we will start with a map f and take the quotient by the equivalence relation $x \sim y \iff f(x) = f(y)$.

There is one last construction that we will need for the future. It is simpler, in that it comes from a set-theoretic construction you are likely already familiar with. All the work comes in giving it a group structure.

Definition 3.22. Let G and H be groups. Their Cartesian product

$$G \times H = \{(g, h) : g \in G, h \in H\}$$

inherits a group structure via

$$(g_1, h_1) \cdot (g_2, h_2) = (g_1 \cdot g_2, h_1 \cdot h_2).$$

Exercise 3.23 (unimportant)**.** Check that this indeed defines a group structure on $G \times H$.

Exercise 3.24 (for those with the necessary background)**.** Show that the statement "$\mathbb{Z}_n \times \mathbb{Z}_m$ is isomorphic to $\mathbb{Z}_{mn}$ when $\gcd(n, m) = 1$" is equivalent to the Chinese remainder theorem. We will be able to prove this statement quite easily later: see Exercise 4.5.

Observe that there are "projection maps" $\pi_G : G \times H \to G$ and $\pi_H : G \times H \to H$ given by forgetting one of the elements of the ordered pair. These are surjective group homomorphisms, and $G \cong (G \times H)/\ker \pi_G$. (Exericse: prove as many of the preceding claims as you like. This will be easier with Theorem 4.3.)

4 Some Classification Theorems

One of the great undertakings of group theory is to classify groups meeting certain conditions. The largest and most famous example is the classification of finite simple groups, which spans decades of work by hundreds of mathematicians. The proof is so long that there is an ongoing project to revise and slim down the presentation: this revised version is still a work-in-progress, but currently spans 10 volumes with an estimated 5 more volumes still to come.

We will, of course, not worry about the classification of finite simple groups. But there are a number of smaller classification theorems that we can understand and extract much value from.

Theorem 4.1 (Lagrange)**.** *Let G be a finite group of order n. Every subgroup of G has order dividing n. In particular, for any $g \in G$, $g^n = 1$.*

Proof. Let H be a subgroup of G, and consider a coset gH. The map of sets $h \mapsto gh$ (called *left-translation* by g) is a bijection between H and gH, since it has inverse $h \mapsto g^{-1}h$. Thus $|gH| = |H|$ for all $g \in G$. But the cosets form a partition of G into disjoint equivalence classes, so $|G| = |H| \cdot [G : H]$, where $[G : H]$ is the number of distinct cosets of H (called the *index* of H in G).

The second claim follows upon considering the cyclic subgroup $\langle g \rangle \subset G$. $\square$

For each $g \in G$, since $g^{|G|} = 1$, we know that there is a minimal positive integer k such that $g^k = 1$. We call this the **order** of g. Note that the order of g is $|\langle g \rangle|$. Lagrange's theorem says that the order of g divides $|G|$.

Exercise 4.2. Show that a group of order p is cyclic for p prime. (Hint: what are the possible orders of elements?)

The above exercise is our first classification theorem.

Theorem 4.3 (First Isomorphism Theorem)**.** *Let G, H be groups, and let $f : G \to H$ be a group homomorphism. Then $\ker f$ is a normal subgroup of G. Let $\pi : G \to G/\ker f$ be the natural projection onto the quotient. Then there is a unique homomorphism $\tilde{f} : G/\ker f \to H$ with $\tilde{f} \circ \pi = f$, and $\tilde{f}$ is an isomorphism $G/\ker f \cong \mathrm{im} f$. This is summarized in the following diagram.*

$$\begin{tikzcd}
\ker f \arrow[r, hook] & G \arrow[r, "f"] \arrow[d, two heads, "\pi"'] & H \\
& G/\ker f \arrow[r, dashed, "\exists! \tilde{f}"] & \mathrm{im}\, f \arrow[u, hook]
\end{tikzcd}$$

In particular, every group homomorphism identifies its image as a quotient of its domain, and every normal subgroup is the kernel of some group homomorphism, namely the projection onto the quotient by that subgroup.

Proof. The only part of the statement left to check (after the previous section) is that there is a unique isomorphism $\tilde{f}$ with $\tilde{f} \circ \pi = f$. If such an isomorphism were to exist, we would need

$$\tilde{f}(\pi(g)) = f(g)$$

for all $g \in G$. Since $\pi(g)$ is the equivalence class $[g]_\sim$ in $G/\ker f$, this forces us to define

$$\tilde{f}([g]_\sim) = f(g).$$

We need to check that this is well-defined, a group homomorphism, and bijective.

(i) For well-definedness, pick any $g \in G$. We need to check that $f(g) = f(gh)$ for any $h \in \ker f$, so that $\tilde{f}([g]_\sim) = f(g)$ does not depend on choice of representative. But indeed, since $h \in \ker f$, we have

$$f(gh) = f(g)f(h) = f(g)$$

as desired.

(ii) To check that $\tilde{f}$ is a homomorphism, compute

$$\tilde{f}([g_1]_\sim \cdot [g_2]_\sim) = \tilde{f}([g_1 g_2]_\sim) = f(g_1 g_2) = f(g_1)f(g_2) = \tilde{f}([g_1]_\sim) \cdot \tilde{f}([g_2]_\sim)$$

as desired.

(iii) Finally, we check that $\tilde{f}$ is bijective. Surjectivity is immediate: for any $y \in \operatorname{im} f$, let $x \in G$ be an element with $f(x) = y$. Then $\tilde{f}([x]_\sim) = f(x) = y$. It remains to check injectivity.

Heuristically, we are collapsing all of $\ker f$ to a single element $[1]_\sim$, the identity coset, and hence "injectivizing" f. This is borne out in computation in the sense that, if $[g]_\sim \in \ker \tilde{f}$, then $\tilde{f}([g]_\sim) = f(g) = 1_H$, so $g \in \ker f$. That is, $g \sim 1$, so $[g]_\sim = [1]_\sim$ is the identity in $G/\ker f$. Thus $\tilde{f}$ is injective, so bijective.

□

The first isomorphism theorem is a great way to construct maps out of quotients. It is usually used according to the following template: if you want to understand G/H, construct a map $G \to K$ whose kernel is H. Then compute its image, which is isomorphic to G/H by the theorem. Try it out on the following long-promised example.

Exercise 4.4. Think of S^1 as the unit circle in $\mathbb{C}^\times$. Show that there is a group isomorphism $S^1 \cong \mathbb{R}/\mathbb{Z}$. (Hint: $e^{i\theta}$. See also Example 3.12.)

Exercise 4.5 (c.f. Exercise 3.24)**.** Prove the Chinese remainder theorem: if $\gcd(n, m) = 1$, then

$$\mathbb{Z}_{mn} \cong \mathbb{Z}_n \times \mathbb{Z}_m.$$

Theorem 4.6 (Cayley)**.** *Every finite group is isomorphic to a subgroup of S_n for appropriate n.*

Cayley's theorem permits you to think of *any* group as a group of bijective functions with the operation given by composition. The idea of the proof is to let $g \in G$ act as a permutation of the elements of G by multiplication on the left. That is, $\sigma_g : h \mapsto gh$ is a permutation of G, so can be understood as an element of $S_{|G|}$.

Exercise 4.7. Check that $g \mapsto \sigma_g$ is an embedding (i.e. injective group homomorphism) of G into $S_{|G|}$. Deduce Cayley's theorem.

Warning: Cayley's theorem does not tell you that every finite group is well-behaved. On the contrary, it tells you that subgroups of S_n can be monstrously ill-behaved, even though we imagine that we understand S_n quite well. However, for abelian groups, we can do quite a bit better.

Theorem 4.8 (Fundamental Theorem of Finite Abelian Groups)**.** *Every finite abelian group is isomorphic to a product of cyclic groups of prime power order. Explicitly, if G is a finite abelian group of order n, we have*

$$G \cong \prod_i \mathbb{Z}_{p_i^{k_i}} \tag{3}$$

where the p_i are not necessarily distinct primes and $\prod_i p_i^{k_i} = n$.

Exercise 4.9 (not for the faint of heart)**.** Look up and read a proof of this theorem, e.g. in [DM72, Section 4.4].

Theorem 4.10. *Let G be a finite cyclic group of order n. Then every subgroup of G is cyclic, and G has a unique cyclic subgroup of order d for every $d \mid n$.*

Proof. First, let G be a cyclic group generated by $g \in G$, and let $H \subset G$ be a subgroup. If $H \neq \{1\}$ (which is clearly cyclic), then there exists some $g^n \in H$ with $n > 0$, so let m be the minimal natural number such that $g^m \in H$. We claim that g^m generates H.

Suppose that some $g' \in H$ is not a power of g^m. Since G is cyclic, write $g' = g^n$ for appropriate n; our assumption is that $m \nmid n$. Now, by the division algorithm, write $n = qm + r$, with $r = 0$ or $0 < r < m$. Then

$$g^n \cdot (g^{-m})^q = g^r \in H$$

since $g^n \in H$, $g^m \in H$, and subgroups are closed under inversion and multiplication. But we assumed m was the minimal positive integer such that $g^m \in H$, so we cannot have $0 < r < m$, i.e. it must be that $r = 0$. But then $n = qm$, so $m \mid n$. This contradiction shows that $H = \langle g^m \rangle$ is cyclic.

Next, without loss of generality, take $G = \mathbb{Z}_n$. If $d \nmid n$, there is no subgroup of G of order d, by Lagrange. Conversely, if $d \mid n$, then $\frac{n}{d}$ is an integer of order d in $\mathbb{Z}_n$, so $\langle \frac{n}{d} \rangle \subset \mathbb{Z}_n$ is a cyclic subgroup of order d. This shows existence.

For uniqueness, suppose $H \subset \mathbb{Z}_n$ is a cyclic subgroup of order d. Then H has a generator, which is an element of order d. We claim that the elements of order d in $\mathbb{Z}_n$ are precisely $\{k\frac{n}{d} : \gcd(k,d) = 1\}$. Indeed, if $\gcd(k,d) > 1$, then $k\frac{n}{d} \cdot \frac{d}{\gcd(k,d)} \equiv 0 \mod n$, and $\frac{d}{\gcd(k,d)} < d$, so the order of $k\frac{n}{d}$ is strictly less than d. Conversely, if $\gcd(k,d) = 1$, one can simply check that the order of $k\frac{n}{d}$ is d.

Thus, H is generated by some $k\frac{n}{d}$ with $\gcd(k,d) = 1$. But then $k\frac{n}{d} \in \langle \frac{n}{d} \rangle$, so $H = \langle k\frac{n}{d} \rangle \subset \langle \frac{n}{d} \rangle$. Then, since $|H| = |\langle \frac{n}{d} \rangle| = d$, we must have $H = \langle \frac{n}{d} \rangle$. $\square$

5 Duality

Our journey towards Fourier analysis begins with an anachronistic perspective. The central role of the **characters** of a group did not become obvious until much after Fourier's work, but starting with them will make our lives much simpler. These will ultimately be the special basis of functions on G that we will use to build the Fourier transform.

Definition 5.1 (c.f. Remark 7.32)**.** Let G be a group, and S^1 the unit circle

$$S^1 = \{x \in \mathbb{C} : |x| = 1\}$$

under multiplication. A **character** of G is a group homomorphism $G \to S^1$. (When we have a topology on our groups, we will demand that characters are continuous too. For finite abelian groups, this distinction does not matter.)

Though singling out a particular class of maps may seem suspicious, this section is dedicated to showing that this is not such a crazy idea. (For the category lovers: it can be shown that S^1 is the only group H such that $\mathrm{Hom}(-, H)$ satisfies the Duality Theorem 5.13.)

Definition 5.2. The **dual group** $\widehat{G}$ of G is its group of characters. If e_1, e_2 are characters of G, their product is defined in the usual way: $(e_1e_2)(g) = e_1(g)e_2(g)$.

Exercise 5.3 (important details)**.** Check that the product of two characters is a character. Check that $e(x) \equiv 1$ is the identity of $\widehat{G}$, and that $e^{-1}(g) = \overline{e(g)}$ gives the inverse of e.

Exercise 5.4. Show that, if e is a character of G and $g \in G$,

$$e(g^{-1}) = \overline{e(g)}.$$

We will want to have a more hands-on understanding of the characters of some of our favorite groups. In fact, computing some examples of $\widehat{G}$ will justify our choice of notation, using e for a character.

Proposition 5.5. *The dual group of $\mathbb{Z}_n$ is identified with the collection of nth roots of unity, which is identified with $\mathbb{Z}_n$ itself. That is, $\widehat{\mathbb{Z}_n} \cong \mathbb{Z}_n$. We say $\mathbb{Z}_n$ is **self-dual**.*

Proof. Recall that the nth roots of unity $\mu_n \subset \mathbb{C}$ are the collection of roots of the polynomial $z^n - 1$ in $\mathbb{C}$. By the fundamental theorem of algebra, there are exactly n roots of this polynomial, and by e.g. computing $\gcd(z^n - 1, (z^n - 1)') = 1$, the roots are all distinct. It is easy to compute that $\{e^{2\pi ik/n} : 1 \leq k \leq n\}$ are nth roots of unity, and since this specifies n elements, we have

$$\mu_n = \{e^{2\pi ik/n} : 1 \leq k \leq n\}.$$

By identifying $e^{2\pi ik/n}$ with $k \in \mathbb{Z}_n$, we obtain an isomorphism

$$\mu_n \cong \mathbb{Z}_n.$$

Exercise 5.6. Check that the above defines an isomorphism of groups. What is the operation in μ_n?

Suppose e is a character of $\mathbb{Z}_n$. Then

$$e(x)^n = e(nx) = e(0) = 1$$

since e is a homomorphism. That is, $e(x)$ is an nth root of unity for any x. Furthermore,

$$e(x) = e(x \cdot 1) = e(1)^x$$

so e is completely specified by its action on 1. This is a rephrasing of the fact that a homomorphism out of a cyclic group is determined by its action on a generator, which leads us to consider

$$\varphi : \widehat{\mathbb{Z}_n} \to \mu_n, \qquad \varphi(e) = e(1).$$

Exercise 5.7. Check that φ as defined above is a homomorphism of groups.

Surjectivity is immediate, since for any $e^{2\pi ik/n} \in \mu_n$, $e(x) := e^{2\pi ikx/n}$ is a character of $\mathbb{Z}_n$ with $\varphi(e) = e^{2\pi ik/n}$. Injectivity is no harder; if $\varphi(e_1) = \varphi(e_2)$, then $e_1(1) = e_2(1)$. But a homomorphism out of a cyclic group is determined by its action on a generator, i.e. $e_i(x) = e_i(1)^x$, so

$$e_1(x) = e_1(1)^x = e_2(1)^x = e_2(x)$$

for every $x \in \mathbb{Z}_n$. Hence $e_1 = e_2$. □

In light of the fundamental theorem of finite abelian groups, it would be useful to understand how dual groups behave under taking products, since we now understand the duals of the "building blocks" $\mathbb{Z}_n$.

Proposition 5.8. *If G and H are finite abelian groups, $\widehat{G \times H} \cong \widehat{G} \times \widehat{H}$.*

Proof. Identify $(f, g) \in \widehat{G} \times \widehat{H}$ with $(x, y) \mapsto f(x)g(y)$ in $\widehat{G \times H}$. To check that this is injective, it suffices to check that its kernel is trivial. But indeed, if (f, g) maps to the identity $1 \in \widehat{G \times H}$, then for 1_G the identity in G we have $f(1_G)g(y) = 1$ for all $y \in H$, and in particular $g(y) \equiv 1$ since $f(1_G) = 1$. On the other hand, $f(x)g(1_H) = f(x) \equiv 1$, and so f and g are both identically 1 and (f, g) is the identity in $\widehat{G} \times \widehat{H}$.

For surjectivity, if $e \in \widehat{G \times H}$, let $e(1_G, y) = g(y)$ and $e(x, 1_H) = f(x)$. If we have any hope, we expect $e = f(x)g(y)$ to be the image of (f, g) under our proposed isomorphism. And indeed, for any $(x, y) \in G \times H$, we have $e(x, y) = e(x, 1_H) \cdot e(1_G, y) = f(x)g(y)$ since e is a character and $(x, y) = (x, 1_H) \cdot (1_G, y)$ in $G \times H$. □

Exercise 5.9 (for the untrusting reader)**.** Check that the assignment $(f, g) \mapsto f(x)g(y)$ is a well-defined group homomorphism, so that the bijection of the above proof is an isomorphism of groups.

Remark 5.10. The isomorphism of Proposition 5.8 is **natural**. The technical meaning (coming from category theory) is that the diagram below commutes, i.e. taking the top-right path is the same as taking the bottom-left path from $\widehat{G_1 \times H_1}$ to $\widehat{G_2} \times \widehat{H_2}$. (Here $f : G_2 \to G_1$ and $g : H_2 \to H_1$ are two group homomorphisms, and h^* is the function on $\widehat{K}$ given by precomposition $e \mapsto e \circ h$.)

$$\begin{tikzcd}
\widehat{G_1 \times H_1} \arrow[r, "\cong"] \arrow[d, "(f\times g)^*"'] & \widehat{G_1} \times \widehat{H_1} \arrow[d, "f^* \times g^*"] \\
\widehat{G_2 \times H_2} \arrow[r, "\cong"] & \widehat{G_2} \times \widehat{H_2}
\end{tikzcd}$$

On the other hand, "naturality" means something to you as a human – something to do with the fact that we didn't need to make any choices. The diagram above should express to you the fact that we defined the isomorphism for every pair of groups all at once, precisely because we made no special choices. Making the notion of naturality mathematically rigorous was the inspiration for the development of category theory, and it is an idea we will see later.

Most mathematicians use the word "natural" in the intuitive sense, and only if they are especially careful do they check the technical requirements. The categorical definition is so effective at capturing our intuition that this is typically not a problem. But this is an endorsement of category theory, not of the careless mathematician. When in doubt, a safer word is "canonical," which expresses a similar intuitive idea but does not come with the baggage of a technical meaning.

Corollary 5.11. *Let G be a finite abelian group. Then $G \cong \widehat{G}$.*

Proof. By the classification of finite abelian groups, G is isomorphic to a product of cyclic groups. Each one is self-dual by Proposition 5.5, and this property is preserved by taking products of groups by Proposition 5.8.

□

Remark 5.12 (c.f. Remark 7.32)**.** For the dual group of a "continuous group" like S^1 or $\mathbb{R}$, we demand that characters are also continuous. We will make precise what this means when we define a **topological group**.

Once we impose such a requirement, a character of S^1 must be a continuous map which turns addition to multiplication (i.e. is a group homomorphism $S^1 \cong [0, 1) \to \mathbb{C}^\times$), so must be of the form $e(x) = e^{2\pi i \gamma x}$. Since we require

$e(0) = e(1)$ (since "$0 = 1$ in S^1"), we need γ to be an integer. In particular, the dual group of S^1 is identified with $\mathbb{Z}$ via $n \mapsto e(x) = e^{2\pi i n x}$. In $\mathbb{R}$, we have no such restriction, and so $\gamma \mapsto e(x) = e^{2\pi i \gamma x}$ identifies $\widehat{\mathbb{R}} \cong \mathbb{R}$, i.e. $\mathbb{R}$ is self-dual.

Theorem 5.13 (Duality)**.** *Let G be a finite abelian group, and let* $\mathrm{ev} : G \to \widehat{\widehat{G}}$ *be the map sending g to* ev_g *defined by* $\mathrm{ev}_g(\chi) = \chi(g)$*. Then* ev *is a natural isomorphism of groups.*

Exercise 5.14 (important detail)**.** Show that $g \mapsto \mathrm{ev}_g$ is a homomorphism of groups.

Proof of the Duality Theorem. We will start by checking that ev is injective. By multiplying by inverses, it suffices to check that $g \neq \mathrm{id}$ implies $\mathrm{ev}_g \neq \mathrm{ev}_{\mathrm{id}} \equiv 1$, i.e. there exists some character $\chi \in \widehat{G}$ taking a value different from 1 on g. (That is, we want to check that ev has trivial kernel.)

Pick $g \in \bigoplus_{i=1}^n \mathbb{Z}_{p_i^{r_i}}$ with last coordinate $\pi_n(g) \neq 0$ without loss of generality; g always has some nonzero coordinate, since it was chosen not to be the identity.

Suppose $N = \prod_{k<n} e^{2\pi i a_k \pi_k(g)/p_k^{r_k}}$. Then pick a_n such that $e^{2\pi i a_n \pi_n(g)/p_n^{r_n}} \neq N^{-1}$, where $\pi_n(g)$ is the nth component of g. Then

$$\chi : x \mapsto e^{2\pi a_1 \pi_1(x)/p_1^{r_1}} \cdot \ldots \cdot e^{2\pi i a_n \pi_n(x)/p_n^{r_n}}$$

is a character which takes a value different from 1 on g. It only remains to check that such an a_n can be found.

But indeed, the generator character $x \mapsto e^{2\pi i x/p_n^{r_n}}$ distinguishes all inputs, so we can always pick the desired a_n: just take $a_n = 0$ if it works, i.e. if $N \neq 1$, or otherwise $a_n = 1$. Since $\pi_n(g) \neq 0$, these take different values on $\pi_n(g)$ (i.e. $e^{2\pi i \pi_n(g)/p_n^{r_n}} \neq 1$), so they can't both be N^{-1} and one of them will give a product not equal to 1. So ev is injective.

By Corollary 5.11, G and $\widehat{\widehat{G}}$ are finite sets of the same order, so an injection between them is a bijection, and we are done.

□

Remark 5.15. In fact, the duality theorem is true for all "locally compact abelian groups," whatever that means. The proof is strikingly difficult, and we will not worry about it. But the recipe is the same: $\mathrm{ev} : G \to \widehat{\widehat{G}}$ is a natural isomorphism.

Exercise 5.16 (for category-lovers)**.** Let G, H be finite cyclic groups, and let $f : G \to H$ be a group homomorphism. The cyclic hypothesis is not necessary but helps avoid distractions. Observe (or prove!) that f defines a homomorphism $f^* : \widehat{H} \to \widehat{G}$ by $\chi \mapsto \chi \circ f$. Define a map $f^{**} : \widehat{\widehat{G}} \to \widehat{\widehat{H}}$ as follows. An element $g \in \widehat{\widehat{G}}$ is a character $g : \widehat{G} \to S^1$. For an element $\chi \in \widehat{H}$, let $f^{**}(g)(\chi) = g(\chi \circ f)$. Check that the evaluation map ev is a *natural isomorphism*, i.e. that the square

$$\begin{tikzcd} G \arrow[r, "\mathrm{ev}"] \arrow[d, "f"] & \widehat{\widehat{G}} \arrow[d, "f^{**}"] \\ H \arrow[r, "\mathrm{ev}"] & \widehat{\widehat{H}} \end{tikzcd}$$

commutes.

Now, let ϕ_G be the isomorphism $G \cong \widehat{G}$ given by $\phi_G(g) = e^{2\pi i g x/|G|}$ (c.f. Proposition 5.5). Check that ϕ is not a natural isomorphism (more properly, not a *di*natural isomorphism, due to the direction of the arrows), i.e. that the square

$$\begin{tikzcd} G \arrow[r, "\phi_G"] \arrow[d, "f"] & \widehat{G} \\ H \arrow[r, "\phi_H"] & \widehat{H} \arrow[u, "f^*"'] \end{tikzcd}$$

does not commute.

Exercise 5.17 (for those with sufficient categorical background)**.** Show that the duality theorem expresses that $\mathrm{Hom}(-, S^1)$ is an equivalence of categories between the category **FinAb** of finite abelian groups and its opposite category. If you like, check that $\mathrm{ev} : 1 \Rightarrow \widehat{\widehat{(-)}}$ is even the unit of an adjunction which witnesses $\mathrm{Hom}(-, S^1)$ as *self-adjoint*. This explains Proposition 5.8 and gives us much more for free.

6 $L^2(G)$

The most natural setting for Fourier analysis is the vector space $L^2(G)$. The classical explanation of the Fourier transform is that it is "writing a function in an appropriate basis." This is excellent intuition, but it remains to be made precise for us. It will turn out to be literally true as stated when G is finite, but will require some interpretation when G is infinite. This interpretation is all on the linear algebra side, e.g. in finding the appropriate definition of the space of functions $L^2(G)$, and understanding what a basis for an infinite-dimensional

vector space looks like. We put off these considerations and demand $|G| < \infty$ for this section.

Definition 6.1. Let G be a finite abelian group and $\mathbb{F}$ a field. Then $L^2(G;\mathbb{F})$ is the set of functions $f : G \to \mathbb{F}$. We imbue it with the structure of a vector space by pointwise addition and scaling, i.e.

$$(\alpha f)(x) = \alpha f(x)$$

and

$$(f+g)(x) = f(x) + g(x).$$

Remark 6.2. When we are ambivalent about the field $\mathbb{F}$, we will omit it, and default to $\mathbb{C}$. Furthermore, the notation $L^2(G)$ will become clearer later; the 2 is because we want $\sum_{g\in G} |f(g)|^2 < \infty$, but this is no imposition at all for functions on finite groups. It will become a relevant point when we consider more general groups; see Definition 8.27 if you cannot wait.

We know a subset of $L^2(G)$: the characters $\widehat{G}$. The crux of Fourier analysis on finite abelian groups is that $\widehat{G} \subset L^2(G)$ forms an orthogonal basis, and this is the "nice" basis that we will express our functions in.

Proposition 6.3 (Orthogonality)**.** *The characters $\widehat{G} \subset L^2(G)$ form an orthogonal basis. In particular,*

$$\langle e_1, e_2\rangle = \begin{cases} 0 & \text{if } e_1 \neq e_2 \\ |G| & \text{if } e_1 = e_2. \end{cases} \tag{4}$$

Proof. We use a cute trick for computing orthogonality. For any $g_0 \in G$, we have

$$e_1(g_0)\langle e_1, e_2\rangle = e_1(g_0)\sum_{g\in G} e_1(g)\overline{e_2(g)}$$

$$= \sum_{g\in G} e_1(g_0)e_1(g)\overline{e_2(g)} = \sum_{g\in G} e_1(g_0 g)\overline{e_2(g)}$$

$$= \sum_{g\in G} e_1(g)\overline{e_2(g_0^{-1}g)}$$

$$= \sum_{g\in G} e_1(g)e_2(g_0)\overline{e_2(g)}$$

$$= e_2(g_0)\sum_{g\in G} e_1(g)\overline{e_2(g)} = e_2(g_0)\langle e_1, e_2\rangle$$

since a sum over $g \in G$ is the same as a sum over $g_0^{-1}g$, and e_1, e_2 are characters, hence $e_2(g_0) = \overline{e_2(g_0^{-1})}$ by Exercise 5.4. Thus we obtain

$$e_1(g_0)\langle e_1, e_2\rangle = e_2(g_0)\langle e_1, e_2\rangle$$

for every $g_0 \in G$, so that either $e_1 = e_2$ or $\langle e_1, e_2\rangle = 0$. In the case that $e_1 = e_2 = e$, we compute

$$\langle e, e\rangle = \sum_{g\in G} |e(g)|^2 = \sum_{g\in G} 1 = |G|.$$

Orthogonal vectors are linearly independent: if $c_1e_1 + ... + c_ne_n = 0$, then

$$0 = \langle c_1e_1 + ... + c_ne_n, e_i\rangle = c_i\langle e_i, e_i\rangle$$

which forces $c_i = 0$ for each i. (Side note: in functional analysis, it is useful to observe that a family $\{e_i\}$ of vectors forms an orthogonal basis if and only if the only vector orthogonal to each e_i is the zero vector.)

Now, we will use finiteness of G quite explicitly. Note that, by Proposition 5.5, $|\widehat{G}| = |G|$. Enumerate the elements of G as $\{g_i : 1 \leq i \leq |G|\}$. The standard basis of $L^2(G)$ consisting of functions f_i with

$$f_i(g_j) = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

certainly has $|G|$ elements, so $L^2(G)$ is $|G|$-dimensional. So $\widehat{G}$ is a linearly independent set of $\dim L^2(G) = |G|$ vectors, so it is a basis.

□

Exercise 6.4. This exercise will show that orthogonality of the characters of $\mathbb{Z}_n$ is equivalent to an already-familiar fact.

Show first that, for characters $e_1, e_2 \in \widehat{G}$, we have $\langle e_1, e_2\rangle = \langle e_1e_2^{-1}, 1\rangle$. In particular, $\widehat{G}$ is an orthogonal set if and only if every non-identity character is orthogonal to the identity character. Now, use the polynomial relation

$$\frac{z^n - 1}{z - 1} = z^{n-1} + z^{n-2} + ... + z + 1$$

to deduce that the characters of $\mathbb{Z}_n$ are orthogonal and have norm $n = |\mathbb{Z}_n|$.

A continuation for those comfortable with tensor products: show that

$$L^2(G \times H) \cong L^2(G) \otimes L^2(H)$$

with inner product $\langle a \otimes b, c \otimes d\rangle = \langle a, c\rangle \cdot \langle b, d\rangle$, and deduce from this that the characters form an orthogonal basis in $L^2(G)$ for any finite abelian group G. (This is much more complicated than the above direct argument, but enlightening in view of our strategy for proving $\widehat{G} \cong G$.)

We now have all we need to define the Fourier transform and bring to life the motivation from the beginning of the section.

Definition 6.5. Let G be a finite abelian group and $f \in L^2(G)$. Then, for any character $e \in \widehat{G}$, the **Fourier coefficient** is

$$\hat{f}(e) = \frac{1}{|G|}\langle f, e\rangle = \frac{1}{|G|}\sum_{g\in G} f(g)\overline{e(g)}. \tag{5}$$

The function $e \mapsto \hat{f}(e)$ describes a function $\hat{f} \in L^2(\widehat{G})$.

If you are so inclined, you can check that $f \mapsto \hat{f}$ is a linear transformation. From this perspective, Theorem 6.6 will show that it is an isomorphism $L^2(G) \cong L^2(\widehat{G})$, and Theorem 6.8 will show that it is *almost* an isometry, up to this pesky factor of $|G|$ which will be addressed in Remarks 6.7 and 8.25.

Theorem 6.6 (Fourier Inversion)**.** *Let G be a finite abelian group and $f \in L^2(G)$. With the Fourier coefficients as before, we have*

$$f(g) = \sum_{e\in\widehat{G}} \hat{f}(e)e(g) \tag{6}$$

for any $g \in G$.

Proof. For any orthonormal basis $\beta = \{\beta_1, ..., \beta_n\}$ of any finite-dimensional vector space V, and any $v \in V$, we have

$$v = \sum_{i=1}^{n}\langle v, \beta_i\rangle.$$

Indeed, since β is a basis, there are some coefficients c_i such that

$$v = c_1\beta_1 + ... + c_n\beta_n.$$

If we take the inner product with some $\beta_j \in \beta$, we obtain

$$\langle v, \beta_j\rangle = \langle\sum_{i=1}^{n} c_i\beta_i, \beta_j\rangle = \sum_{i=1}^{n} c_i\langle\beta_i, \beta_j\rangle$$

and since $\langle\beta_i, \beta_j\rangle = \delta_{ij}$, only the $i = j$ term is nonzero, hence

$$\langle v, \beta_j\rangle = c_j\langle\beta_j, \beta_j\rangle = c_j.$$

Thus the inner product extracts the coefficients in the linear combination; this justifies calling $\langle\cdot, \beta_j\rangle$ the projection onto β_j when β forms an orthonormal basis.

In our particular case, the characters are not quite orthonormal, but they are orthogonal, and $||e||^2 = |G|$. Consequently, when $f = \sum_{e \in \widehat{G}} c_e e$ for some coefficients c_e, we have $\langle f, e \rangle = |G| c_e$, and thus $\frac{1}{|G|}\langle f, e \rangle$ extracts the desired coefficients. Once again, it is profitable to think of $\hat{f}(e) = \frac{1}{|G|}\langle f, e \rangle$ as the projection of f onto the character e. $\square$

Remark 6.7. Here and elsewhere, the factor of $|G|$ arising from $\langle e, e \rangle = |G|$ is a thorn in our side. We could remove it by scaling the inner product itself, i.e. defining

$$\langle f, g \rangle' = \frac{1}{|G|} \sum_{x \in G} f(x)\overline{g(x)}.$$

But this hardly solves our problem, as we are carrying around a factor of $\frac{1}{|G|}$ in our sums now. We will see later that this is a symptom of our choice of *measure* on G.

Theorem 6.8 (Plancherel)**.** *For $f \in L^2(G)$, we have*

$$||f||^2 = \sum_{g \in G} |f(g)|^2 = |G| \sum_{e \in \widehat{G}} |\hat{f}(e)|^2 = |G| \cdot ||\hat{f}||^2. \tag{7}$$

Proof. There is nothing fancy to do – just compute. Using Theorem 6.6, we have

$$f = \sum_{e \in \widehat{G}} \hat{f}(e) e$$

and so

$$||f||^2 = \langle f, f \rangle = \langle \sum_{e \in \widehat{G}} \hat{f}(e) e, \sum_{e \in \widehat{G}} \hat{f}(e) e \rangle.$$

We use a typical trick for orthogonal bases: expanding the above inner product by linearity and conjugate-symmetry yields

$$\langle \sum_{e \in \widehat{G}} \hat{f}(e) e, \sum_{e \in \widehat{G}} \hat{f}(e) e \rangle = \sum_{e_1 \in \widehat{G}} \langle \hat{f}(e_1) e_1, \sum_{e_2 \in \widehat{G}} \hat{f}(e_2) e_2 \rangle = \sum_{e_1 \in \widehat{G}} \sum_{e_2 \in \widehat{G}} \langle \hat{f}(e_1) e_1, \hat{f}(e_2) e_2 \rangle,$$

and since the characters are orthogonal, all the "cross-terms" with $e_1 \neq e_2$ vanish. This leaves us with

$$||f||^2 = \sum_{e \in \widehat{G}} \langle \hat{f}(e) e, \hat{f}(e) e \rangle = \sum_{e \in \widehat{G}} |\hat{f}(e)|^2 \cdot |G| = |G| \cdot ||\hat{f}||^2$$

since the inner product is conjugate-symmetric and $\langle e, e \rangle = |G|$. $\square$

Exercise 6.9 (important for later)**.** Show that, for $f \in L^2(G; \mathbb{C})$, f is real-valued if and only if $\hat{f}(e^{-1}) = \overline{\hat{f}(e)}$ for all $e \in \widehat{G}$. This will be surprisingly useful for constructing examples later. If you define a function by its Fourier transform, it can be hard to imagine what the un-transformed function looks like.

Exercise 6.10 (enlightening but optional)**.** Let $f \in L^2(\mathbb{Z}_n; \mathbb{R})$. Deduce from the previous exercise that Fourier inversion expresses f as an $\mathbb{R}$-linear combination of sines and cosines, and compute the coefficients of these sines and cosines in terms of the Fourier coefficients $\hat{f}$. (As a further exercise, if you like: show that this is invertible, i.e. that from a sum of sines and cosines one can recover the Fourier coefficients. Conclude that these sines and cosines form a basis for $L^2(\mathbb{Z}_n; \mathbb{R})$.)

This exercise is particularly useful if you are averse to complex numbers and want to recover some of the Fourier apparatus in $L^2(\mathbb{Z}_n; \mathbb{R})$. This is analogous to the situation in $L^2(S^1)$, which you may have seen. After we see the Fourier transform in more generality, feel free to redo this exercise with S^1 in place of $\mathbb{Z}_n$.

As hinted above, analogues of Fourier inversion and Plancherel hold for $L^2(S^1)$ and $L^2(\mathbb{R})$. But they are much more difficult in these settings, because these are infinite-dimensional spaces. Indeed, we relied on finite-dimensionality quite heavily. (If you feel so inclined, you should go back and check where we used this assumption.)

Here are two nice exercises to force your mind to be flexible in applying the theorems of this section.

Exercise 6.11. Show that, if $H \subset G$ is a subgroup and $e \in \widehat{G}$, then

$$\sum_{h \in H} e(h) = \begin{cases} 0 & \text{if } e|_H \neq 1 \\ |H| & \text{if } e|_H = 1. \end{cases}$$

(Hint: think of the sum as $\langle e, 1 \rangle$ in $L^2(H)$.)

Exercise 6.12. Show that, for $g_1, g_2 \in G$, we have

$$\sum_{e \in \widehat{G}} e(g_1)\overline{e(g_2)} = \begin{cases} 0 & \text{if } g_1 \neq g_2 \\ |G| & \text{if } g_1 = g_2. \end{cases}$$

(Hint: think of g_1, g_2 as elements of $\widehat{\widehat{G}}$ via the isomorphism ev.)

As a nice application, try the following.

Exercise 6.13. Prove Fourier inversion, Theorem 6.6, by brute force: compute $\sum_{e \in \widehat{G}} \hat{f}(e)e(g)$ directly and check that it equals $f(g)$.

(Hint: expand what $\hat{f}(e)$ means in that sum, and switch the order of summation. At some point, you will want to evaluate a sum of the form given in Exercise 6.12.)

7 Interlude: Topology

Those with some analysis experience will recall that the real line comes with a notion of an *open* subset: one that contains an open interval around each one of its points. Proposition 7.16 will show that continuity of a function $f : \mathbb{R} \to \mathbb{R}$ can be expressed entirely in terms of open sets and their preimages under f. This leads us to reverse the machine, in the following sense. If X is merely a set, we can *declare* certain subsets to be open by divine decree, and as long as the open subsets satisfy some collection of rules, we obtain a notion of continuity for functions on X. What should those rules be? They appear in Definition 7.1 below, which axiomatizes the important properties of open sets in $\mathbb{R}$. You should mentally compare each axiom to the behavior of open sets in $\mathbb{R}$ as you read.

Definition 7.1. Let X be a set. A *topology* on X is a family T of subsets of X (i.e. an element of T is a subset $U \subset X$) satisfying the following axioms:

(T1) $X \in T, \emptyset \in T$.

(T2) If $\{U_\alpha\}$ is a (possibly uncountable) collection of elements of T, then $\bigcup_\alpha U_\alpha$ is in T.

(T3) If $U_1, U_2 \in T$, then $U_1 \cap U_2 \in T$. By induction, finite intersections of elements of T are in T.

We call elements of T the **open sets** of X.

Example 7.2. Our favorite subsets of $\mathbb{R}$ are the *open intervals* (a, b) with $a, b \in \mathbb{R} \cup \{\pm\infty\}$. A subset of $\mathbb{R}$ is open if it is a union (possibly uncountable) of open intervals. Equivalently, a subset $U \subset \mathbb{R}$ is open if for every $x \in U$, there is some open interval $(x - \delta, x + \delta) \subset U$.

Exercise 7.3. Check that these open sets satisfy the axioms for a topology on $\mathbb{R}$.

As a final note, observe that

$$\bigcap_{n \geq 1} (a - \frac{1}{n}, a + \frac{1}{n}) = \{a\}$$

is not open, so infinite intersections of open sets need not be open.

Remark 7.4. Because every open set in $\mathbb{R}$ is a union of open intervals, we say that the open intervals are a **base** for the topology on $\mathbb{R}$, or that they **generate** that topology. If a family of sets generates a topology with unions and intersections, rather than merely unions, we call it a **subbase** for that topology. (By

"$\{U_\alpha\}$ generates the topology T," we mean that T is the smallest collection of subsets of X containing all the U_α and satisfying the axioms (T1)-(T3).)

Example 7.5. We declare every subset of $\mathbb{Z}$ and $\mathbb{Z}_n$ to be open. Check that this satisfies the axioms of a topology. We call it the **discrete topology**. On the other hand, we have the **indiscrete topology**, where we say that only X and $\emptyset$ are open.

Definition 7.6. Let X be a space with a topology T. (We call the pair (X, T) a **topological space**, and we omit the topology T when it is clear.) A **closed set** in X is a set whose complement is open. The **closure** of a subset $E \subset X$ is the smallest closed set containing E, denoted $\overline{E}$.

Exercise 7.7. Check that every set has a closure. (Hint: by (T2) and de Morgan's laws, an arbitrary *intersection* of closed sets is closed. So define the closure of E to be the intersection of all closed sets containing E. Why is this nonempty, and why does it satisfy the advertised definition?)

Definition 7.8. Let X be a topological space. A set $A \subset X$ is **dense** if its closure $\overline{A} = X$.

Exercise 7.9 (for those with preconceived notions)**.** Check that this definition aligns with whatever notion of density you already have in your mind. For instance, check that $A \subset X$ is dense if and only if every nonempty open subset of X intersects A nontrivially. Furthermore, for $X = \mathbb{R}$, check that $A \subset \mathbb{R}$ is dense if and only if every $x \in \mathbb{R}$ is the limit of a sequence of elements of A.

(This last part is tricky without further real analysis experience. Perhaps it would be useful to check that the limit of a sequence is a point x such that, for every open neighborhood U of x, all but finitely many terms of the sequence lie in U. Then apply the first part of the exercise to conclude the second part by considering shrinking open intervals around a point.)

Remark 7.10. The problem with any introduction to topology is that one has to develop a visual intuition without relying too heavily on it. For instance, picturing every open set as an interval in $\mathbb{R}$ is too restrictive. Broadly speaking, an open set is an "open ball," i.e. a collection of nearby points which "does not contain its boundary," in the sense that there is no point furthest from the center of the ball. (You can picture a disk in the plane $\mathbb{R}^2$ with its boundary, the circle, excluded.)

A closed set, on the other hand, is a set which contains its boundary. (This is literally true once you define the topological boundary of a set.) You should

picture a closed set as a disk which contains its boundary circle.

Definition 7.11. A topological space X is called **Hausdorff** if, for every pair of points $x \neq y \in X$, there exist open sets $U \ni x$ and $V \ni y$ with $U \cap V = \emptyset$.

The Hausdorff condition is one in a hierarchy of "separation axioms" that are remarkably useful for technical arguments. It is extremely scary but surprisingly common that two points fail to be "topologically distinguishable," i.e. have distinct collections of open neighborhoods. Luckily for us all, any reasonably-behaved space is Hausdorff. (Unless you are an algebraic geometer, for whom Hausdorffness is alarming; see Remark 7.25.)

Exercise 7.12 (extremely useful)**.** Show that, in a Hausdorff topological space, points are closed. That is, if $x \in X$, the set $\{x\}$ is closed.

Exercise 7.13 (easy and important)**.** Show that $\mathbb{R}$ is Hausdorff.

Now we are ready for the payoff, as promised in the introduction to the section. The data of a topological space allows us to express the notion of continuity without reference to distances, norms, or anything of the sort.

Definition 7.14. Let $f : X \to Y$ be a function between topological spaces. We say f is **continuous** if $f^{-1}(U)$ is open in X whenever U is open in Y.

Remark 7.15. Note carefully the use of $f^{-1}(U)$ ("preimage") rather than the "forward image" $f(U)$. If U open implies $f(U)$ open, we call f an *open function*. There are numerous good reasons for using preimages for continuity. One of them is that it is given to us by nature: we have a notion of continuity on $\mathbb{R}$, and this is the equivalent condition, as we will prove shortly. Another is that preimages behave much more reasonably with respect to set operations than forward images. If you like, you can check that $f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B)$, etc. for any set operation you can think of. It is usually not true that $f(A \cap B) = f(A) \cap f(B)$, etc.

Proposition 7.16 (better as an exercise, if you can stomach it)**.** *Continuity in the sense of Definition 7.14 is equivalent to the usual formulation: $f : \mathbb{R} \to \mathbb{R}$ is continuous if, given $\epsilon > 0$, there exists $\delta > 0$ such that $|x - y| < \delta$ implies $|f(x) - f(y)| < \epsilon$.*

Proof. Suppose first that f is continuous in the usual ϵ-δ sense, and suppose $U \subset \mathbb{R}$ is open. Fix an $x \in f^{-1}(U)$. We want to show that there is an open

interval containing x and contained in $f^{-1}(U)$, so that $f^{-1}(U)$ contains such a neighborhood of each of its points, and is thus open by definition of the topology on $\mathbb{R}$.

Since $x \in f^{-1}(U)$, we have $f(x) \in U$. Since U is open, there is an open interval (often called an **open ball**) $B_r(f(x)) := (f(x) - r, f(x) + r)$ contained in U, by definition of the topology on $\mathbb{R}$. Observe that $|f(x) - f(y)| < r$ means $f(y) \in B_r(f(x)) \subset U$. But then, since f is ϵ-δ continuous, take $\epsilon = r$ and obtain δ such that, for every $|x - y| < \delta$, we have $|f(x) - f(y)| < r$.

Rephrasing this again, $|x - y| < \delta$ means $y \in B_\delta(x)$, so our statement about continuity means that every $y \in B_\delta(x)$ has $f(y) \in B_r(f(x)) \subset U$, i.e. $y \in f^{-1}(U)$. Thus $B_\delta(x) \subset f^{-1}(U)$. But $B_\delta(x)$ is an open interval containing x, so since x was arbitrary, we have shown that $f^{-1}(U)$ is open, and hence f is continuous in the topological sense.

Conversely, suppose f is continuous in the topological sense. Fix some $x \in \mathbb{R}$, and let $\epsilon > 0$. We want to find a δ to satisfy the ϵ-δ definition of continuity. To this end, consider the open ball $B_\epsilon(f(x)) = (f(x) - \epsilon, f(x) + \epsilon)$, which is open in $\mathbb{R}$. Then $f^{-1}(B_\epsilon(f(x)))$ is open by topological continuity, so it contains some open interval around each of its points. In particular, it contains an interval around $x \in f^{-1}(B_\epsilon(f(x)))$. By shrinking the interval, we might as well assume it is centered at x, i.e. $(x-\delta, x+\delta) = B_\delta(x) \subset f^{-1}(B_\epsilon(f(x)))$. But, similarly to before, this is a rephrasing of the fact that $|x-y| < \delta$ implies $|f(x) - f(y)| < \epsilon$, i.e. $y \in B_\delta(x)$ implies $f(y) \in B_\epsilon(f(x))$. So f is continuous in the ϵ-δ sense. $\square$

Exercise 7.17. Show that a function $f : X \to Y$ between topological spaces is continuous if and only if $f^{-1}(K)$ is closed whenever K is closed. Deduce that, if $f : X \to \mathbb{R}$ is continuous, the **curve** $C = \{x : f(x) = 0\} = f^{-1}(\{0\})$ is closed. (You should picture a curve or hypersurface in $\mathbb{R}^n$, which is defined by the vanishing of some continuous function. Hint: points are closed in $\mathbb{R}$, Exercises 7.12 and 7.13.)

Definition 7.18. Let X, Y be topological spaces. A function $f : X \to Y$ is a **homeomorphism** if it is a continuous bijection, and f^{-1} is also continuous.

Remark 7.19. Two comments: firstly, the name "homeomorphism" is an egregious historical relic which we cannot shake. It is the appropriate notion of isomorphism for topological spaces: it tells you that, after relabeling $x \in X$ by $f(x) \in Y$, the topology is the same. Secondly, the requirement that f^{-1} be continuous is not automatically satisfied: can you find a counterexample? (As Exercise 7.39 shows, this is tricky. You will have to consider some uncomfortable spaces.)

Given a topological space, we would like to be able to build related spaces and have them inherit a topology. For instance, if X and Y are topological spaces, we would like to give $X \times Y$ a topology in some compatible way. The following two ideas will help us do this in a systematic way, and will produce many useful examples.

Definition 7.20. Suppose X is a set with functions of sets $f_i : X \to Y_i$ for some indices i (maybe infinitely many). If each Y_i already has a topology, then the **initial topology** on X with respect to the family $\{(f_i, Y_i)\}$ is the smallest topology on X (i.e. the one with fewest open sets; some authors say **coarsest**) such that each f_i is continuous. Explicitly, the topology is generated by sets of the form $f_i^{-1}(U)$ for U open in Y_i. (If there is only one map f_0, then the $f_0^{-1}(U)$ are all the open sets already. Why?)

Definition 7.21. Suppose once again that X is merely a set, and we have a family of topological spaces $\{Y_i\}$ with maps $f_i : Y_i \to X$ (note the direction of the arrows). Then the **final topology** on X with respect to this family of maps is the largest topology (i.e. the one with the most open sets; some authors say **finest**) such that each f_i is continuous. Explicitly, $U \subset X$ is open if and only if $f_i^{-1}(U)$ is open in Y_i.

The words "initial" and "final" are category theory words, and indeed these constructions can be expressed categorically. However, the idea is surprisingly complicated, and we will not worry about it.

Example 7.22. Let X be a topological space and $Y \subset X$ a subset. There is an inclusion map $\iota : Y \hookrightarrow X$. We give Y the initial topology with respect to this map, which we call the **subspace topology**. Since $\iota^{-1}(U) = U \cap Y$, this means that a subset of Y is open if and only if it is the intersection of some open set in X with Y.

Example 7.23. Let X and Y be topological spaces. We want to topologize their Cartesian product $X \times Y$. To do this, observe that $X \times Y$ comes with natural maps $\pi_X : X \times Y \to X$ and $\pi_Y : X \times Y \to Y$ given by projecting onto one factor. We give $X \times Y$ the initial topology with respect to these projections, which we call the **product topology**. The product topology is generated by **open rectangles** $U \times V$ with $U \subset X$ open and $V \subset Y$ open.

Exercise 7.24 (amazingly enlightening for algebraic geometers)**.** For any topological space X, we have a special subspace of the product $X \times X$ given by $\Delta := \{(x, x) : x \in X\}$. This subspace Δ is called the *diagonal*. Show that a space X is Hausdorff if and only if the diagonal is closed in $X \times X$.

Remark 7.25. In algebraic geometry, Hausdorffness is the "wrong" separation condition, as in reasonable ("Noetherian") cases it is *open* sets which are compact in the Zariski topology of a variety. Instead, the analogous notion is *separatedness*, which is precisely the condition that the diagonal is closed. How is it different from Hausdorffness, then? Answer: The product of varieties $X \times Y$ is not the product of topological spaces, i.e. does not carry the product topology. (Do not be disturbed: it is just the categorical product of the varieties, which happens to carry a different topology. This is a glimpse of the fact that varieties carry much more structure than merely a topology.)

Example 7.26. Suppose $f : X \to Y$ is surjective and X already has a topology. The final topology on Y induced by f is called the **quotient topology**, and it says that a set $V \subset Y$ is open if and only if $f^{-1}(V)$ is open in X. An important special case (which is really the only case there is; see Exercise 7.27) is the following. Suppose X is a topological space and $\sim$ is an equivalence relation on X. Then there is a natural map $q : X \to X/\sim$, called the projection onto the quotient by $\sim$, and it is clearly surjective.

If f is a surjection $X \to Y$ and Y has the quotient topology for this map, we call f a **quotient map**. This distinction is useful when we already have topologies on X and Y, and independently construct a map f which happens to be a quotient map. As remarked already, f is a quotient map exactly when a set U is open if and only if $f^{-1}(U)$ is open.

Exercise 7.27 (an enlightening peek into universal algebra)**.** Suppose $f : X \to Y$ is a surjective function of sets, and X is a topological space. Define $\ker f$ to be the equivalence relation where $x \sim y$ if and only if $f(x) = f(y)$.

(a) Show that $\ker f$ really is an equivalence relation.

(b) Show that we have a first isomorphism theorem: the map $f : X \to Y$ factors through the natural projection $q : X \to X/\ker f$, i.e. there is a unique map $\tilde{f}$ making the following diagram commute.

$$\begin{tikzcd} X \arrow[r, two heads, "f"] \arrow[d, "q"'] & Y \\ X/\ker f \arrow[ur, dashed, "\exists!\tilde{f}"'] & \end{tikzcd}$$

(c) Show that, if we give $X/\ker f$ the quotient topology for the natural projection, and we give Y the quotient topology for f, the map $\tilde{f}$ is a homeomorphism.

(d) Conclude that, if Y has the quotient topology for a surjection $f : X \to Y$, then Y is homeomorphic to $X/\ker f$ with its quotient topology for the natural projection.

Warning: the above topologies are so natural that we will put them on subsets, products, and quotients without comment.

We now have natural choices of topology for many of the sorts of spaces we will consider. We can begin to put them to use in thinking about groups.

Definition 7.28. Let G be a group. A topology on G makes it into a *topological group* if $i : g \mapsto g^{-1}$ and $m : (g, h) \mapsto gh$ are continuous.

Example 7.29. The real line $\mathbb{R}$ and the circle S^1 are topological groups with their usual topologies. (What is the topology on S^1? Answer: it is either the subspace topology for $S^1 \subset \mathbb{R}^2$, or else the quotient topology for $S^1 \cong \mathbb{R}/\mathbb{Z}$.)

Exercise 7.30 (mandatory)**.** For the moment, think of S^1 as the unit circle in $\mathbb{C}^\times$. Show that your isomorphism of groups from Exercise 4.4 is in fact a homeomorphism of topological groups $S^1 \cong \mathbb{R}/\mathbb{Z}$.

Example 7.31. Let G be a topological group. We give a subgroup $H \subset G$ the subspace topology. If $\pi : G \to G/H$ is the natural projection onto the quotient, we give G/H the quotient topology. If K is another group, we give the direct product $G \times K$ the product topology. If you are feeling pedantic, check that each of these topologies makes the new group into a topological group.

Remark 7.32. If G is a topological group, we (re)define $\widehat{G}$ to be the set of *continuous* characters of G. (When G is finite, we put the discrete topology on it, so this definition is no different from our earlier one.)

Before we continue with groups, here are two ideas which fall under the purview of "point-set topology." The first is more easily understood than the second. They will both be important for us to understand which groups behave well enough to work with.

Definition 7.33. Let X be a topological space. X is said to be **disconnected** if there exist open sets $U, V \subset X$ which are disjoint, i.e. $U \cap V = \emptyset$, and cover X, i.e. $U \cup V = X$. If X is not disconnected, we (quite reasonably) say X is **connected**.

It is sometimes useful to think about **connected components** of a space, i.e. maximal connected subsets.

Exercise 7.34 (easy but enlightening)**.** Call a set $E \subset X$ **clopen** if E is both closed and open in X. Show that a topological space X is connected if and only if the only clopen subsets of X are $\emptyset$ and X itself.

Secondly, let us discuss a particularly important topological idea which is somewhat unapproachable, but in fact quite nice.

Definition 7.35. Let X be a topological space. We say X is **compact** if, for every family of open sets $\{U_\alpha\}$ with $\bigcup_\alpha U_\alpha = X$, there exists a finite subfamily $U_1, ..., U_n$ such that $\bigcup_{i=1}^n U_i = X$. That is, "every open cover of X has a finite subcover."

To give you a feel for using the definition of compactness, here are a couple of point-set topology exercises.

Exercise 7.36. Suppose X is a compact topological space and $K \subset X$ is closed. Show that K is compact.

Exercise 7.37. Suppose that X is Hausdorff and $K \subset X$ is compact. Show that K is closed.

As a matter of fact, we have now met the only two topological ideas which you should ever imagine to be preserved by forward images.

Exercise 7.38. Let $f : X \to Y$ be a continuous function, and let $K \subset X$ be a subset.

(a) Show that, if K is compact, then $f(K)$ is compact. (The special case of a function $\mathbb{R} \to \mathbb{R}$ and $K = [a, b]$ is the Extreme Value Theorem.)

(b) Show that, if K is connected, then $f(K)$ is connected.

Let us put the preceding exercises to good use. Here is a useful criterion for a map being a homeomorphism, which you should prove if you intend to ever use.

Exercise 7.39. Let $f : X \to Y$ be a continuous bijection of topological spaces. If X is compact and Y is Hausdorff, show that f is a homeomorphism. (This is not as scary as it appears. It is essentially a shuffling of definitions, once you have the previous exercises down.)

With that experience under your belt, here is a criterion for compactness in $\mathbb{R}^n$ which will save you a lot of headaches. It also possibly defends the use of the name "compact" in your mind.

Theorem 7.40 (Heine-Borel)**.** *A subset of $\mathbb{R}^n$ is compact if and only if it is closed and bounded.*

Exercise 7.41 (how to think about compactness)**.** Compactness is a tricky idea, and much ink has been spilled about how to teach it. One school of thought says that it simply is what it is, and students should become comfortable with it by working through exercises. Another says that it should be understood as the thing which is characterized in various "nice" spaces by various "nice" theorems, like Heine-Borel or Arzelá-Ascoli. A third school of thought is contained in this exercise. In essence, it says that "compactness is one half of finiteness."

Let X be a topological space. Show that X is compact and discrete (i.e. has the discrete topology) if and only if X is finite and Hausdorff. You may also consider reading this MathStackExchange post. It will not help you with the exercise, except perhaps in motivation.

Definition 7.42. Let X be a topological space. X is **locally compact** if, for every point $x \in X$, there is some open set U and some compact set K with $x \in U \subset K$.

This is a technical definition, expressing the idea that "every point has a compact neighborhood" while avoiding the conflict that you imagine a compact set to be closed but a neighborhood to be open. We will not stress about the local compactness hypothesis which we will carry around for the rest of these notes; its primary purpose is to make Theorem 8.22 true.

Definition 7.43. A group G is a **locally compact abelian group** (often abbreviated **LCA group**) if it is a topological group which is abelian, locally compact, and Hausdorff. (This last requirement could be omitted, but it is amazingly convenient and not very restrictive.)

LCA groups will be the natural setting for all of our future ideas, owing mostly to the existence of a *Haar measure*, Theorem 8.22. We will be able to integrate functions on LCA groups, which will give rise to an inner product space of functions on such groups, bringing us closer to the idea of the Fourier transform as "just picking a nice basis."

Example 7.44. Our favorite groups $\mathbb{R}, S^1, \mathbb{Z}$, and $\mathbb{Z}_n$ are all LCA groups. If you are so inclined, you should check this statement.

Remark 7.45 (for those with topological background)**.** The point is that a manifold is a locally compact Hausdorff space, and the important examples of LCA groups come from manifolds. For instance, $\mathbb{R}$ and S^1 are 1-manifolds, and $\mathbb{Z}$ and $\mathbb{Z}_n$ are 0-manifolds, whatever that means.

Exercise 7.46 (tricky)**.** Show that $\mathbb{Q}$ with the subspace topology from $\mathbb{Q} \subset \mathbb{R}$ is not locally compact (and hence not a manifold, by the preceding remark).

Exercise 7.47 (easy)**.** Check that $\mathbb{Z}$ is not compact but $\mathbb{Z}_n$ is.

There is one last technical point that you may not have considered. We need a topology on $\widehat{G}$, given one on G. We want a construction that works for any LCA group G, and satisfies three convenient properties:

(i) If G is self-dual (e.g. $G = \mathbb{Z}_n$ or $G = \mathbb{R}$), we want the isomorphism $G \cong \widehat{G}$ to be a homeomorphism.

(ii) We want the duality isomorphism $G \cong \widehat{\widehat{G}}$ to be a homeomorphism. (We only know that this isomorphism exists at all for finite abelian groups G. You can imagine what the map is in general, but proving that it is a group isomorphism is remarkably difficult.)

(iii) If G is an LCA group, we want $\widehat{G}$ to be LCA too.

There is such a topology, called the **compact-open topology**. You should think of it as the thing which satisfies the demands above. But here is a formal definition.

Definition 7.48. Let X, Y be topological spaces and $C(X, Y)$ the set of continuous functions $X \to Y$. The **compact-open topology** on $C(X, Y)$ is the topology generated by $V(K, U) = \{f \in C(X, Y) : f(K) \subset U\}$ where $K \subset X$ is compact and $U \subset Y$ is open. That is, functions which send the same compact set into the same open set are close together.

Since $\widehat{G} \subset C(X, Y)$ (Remark 7.32), we give $\widehat{G}$ the subspace topology and call it the compact-open topology too.

Exercise 7.49 (not for a first read-through)**.** Show that the compact-open topology on $\widehat{G}$ achieves objectives (i)-(iii) from above. (This is not for the faint of heart; especially (iii).)

Remark 7.50 (for those with sufficient algebraic topology background)**.** The compact-open topology arises naturally in another setting as a tool for putting

the "right" topology on a space of functions. We like to think of a homotopy $H : X \times [0,1] \to Y$ as the same as a path in $\mathrm{Hom}(X,Y)$, i.e. $H : [0,1] \to \mathrm{Hom}(X,Y)$. If this is to be made precise, we need to topologize the space $\mathrm{Hom}(X,Y)$ so that a continuous map $Z \to \mathrm{Hom}(X,Y)$ is the same as a continuous map $X \times Z \to Y$ (and then take $Z = [0,1]$). The correct topology for the job is once again the compact-open topology.

This discussion can be summarized by saying that the compact-open topology on $\mathrm{Hom}(X,Y)$ makes $\mathrm{Hom}(X,-)$ and $X \times -$ into an adjoint pair of functors on the category of topological spaces. (You should think of this as analogous to the Hom-tensor adjunction in algebraic settings.)

Furthermore, it allows us to say cute things like "the fundamental group is the collection of path components of the space of loops in X" and "the mapping class group is the collection of path components of the space of homeomorphisms $X \to X$."

8 Haar Measure

Like with topology, we are interested in the minimal supporting structure needed to generalize some otherwise interesting concept. Whereas topology comes from a desire to understand continuity, we will now define a *σ-algebra* in order to understand *measure*.

Perhaps the notion of measure is more natural from the perspective of probability theory. There, a subset of the sample space is called an event, and the measure of an event is its probability. Of course, the measure of the entire space is the probability of any outcome, which is 1, and so we call such a space a *finite measure space* ("finite" modifies "measure," not "space") and the measure a *probability measure*.

One useful perspective shift is to think instead of an experiment as a random variable $X : S \to \mathbb{R}$, where S is the sample space, and measure the likelihood of an outcome landing in some subset of $\mathbb{R}$. In this case, we want to define a measure μ on $\mathbb{R}$. We make the eminently reasonable demand that the measure of an interval (a,b) is $b-a$, i.e. its length. Perhaps it is reasonable to expect measures to behave well with respect to taking unions of intervals, but then it is no longer obvious that any subset of $\mathbb{R}$ has a well-defined measure, as different coverings by intervals might give conflicting notions of length.

In fact, our greatest fears come to pass: if we demand that $\mu((a,b)) = b-a$, not every subset of $\mathbb{R}$ is measurable. That is, we cannot define a coherent notion of measure on every subset of $\mathbb{R}$. (Of course, if our expectations were lower, we could measure every subset of $\mathbb{R}$, e.g. by setting $\mu(E) = 0$ for every E.) To remedy this, we need a supporting structure which tells us what subsets of $\mathbb{R}$

we are allowed to measure.

Remark 8.1. In fact, it requires the Axiom of Choice to construct a non-measurable subset of $\mathbb{R}$. If you are so inclined, look up the *Vitali set(s)* for an example.

Definition 8.2. Let X be a set. A σ-algebra on X is a family A of subsets of X (i.e. $A \subset \mathcal{P}(X)$), called the measurable subsets of X, such that

(S1) $X \in A$.

(S2) If $E \in A$, then $E^c \in A$.

(S3) If $E_1, E_2, ...$ is a countable family of sets in A, then their union $\bigcup_{n \geq 1} E_n$ is in A.

This should remind you heavily of the definition of a topology. However, it is stronger in some ways, and weaker in others. For instance, topologies contain all unions, even uncountable ones. On the other hand, they do not typically contain complements of their members; if E and E^c are both open, then the space is disconnected (Exercise 7.34).

Nevertheless, the apparent connection between a σ-algebra and a topology motivates the following definition.

Definition 8.3. Let X be a topological space. The *Borel σ-algebra* on X is the smallest σ-algebra such that the open sets of X are measurable.

Example 8.4. Since the open intervals (a, b) form a base for the topology on $\mathbb{R}$, the Borel σ-algebra on $\mathbb{R}$ is generated by the open intervals.

Just as a topology is a stepping-stone to the notion of continuous function, we will immediately leverage our notion of measurable subsets to define a *measurable function.*

Definition 8.5. Let $f : X \to Y$ be a function between two spaces, each equipped with a σ-algebra. Then f is measurable if, for every measurable $E \subset Y$, $f^{-1}(E) \subset X$ is measurable too.

This mimics the definition of continuity quite clearly. In analogy with that situation, using preimages gives the "right" definition because set operations behave nicely with respect to preimages, but not images. For example, in light of the previous example, a function $f : X \to \mathbb{R}$ is measurable if and only if $f^{-1}((a, b))$ is measurable for every open interval (a, b) (where one of a, b could be $\pm\infty$). For reasons which will become clearer later, it is usually more useful to think of the same fact in the following way.

Exercise 8.6. Let $f : X \to \mathbb{R}$ be a function. Show that f is measurable if and only if $f^{-1}([a, b))$ is measurable for every $a, b \in \mathbb{R} \cup \{\pm\infty\}$. That is, preimages of half-open intervals are measurable. In particular, a random variable on a

sample space X is a measurable function $X \to \mathbb{R}$. (Depending on the context in which you were first exposed to the notion of random variables, this may have already been your definition.)

Finally, we are able to get at the idea that originally motivated us: measure.

Definition 8.7. Let X be a set with σ-algebra A. A function $\mu : A \to \mathbb{R} \cup \{\infty\}$ is a *measure* if

(M1) $\mu(E) \geq 0$ for all $E \in A$.

(M2) $\mu(\emptyset) = 0$.

(M3) If $E_1, E_2, ...$ is a countable family of **disjoint** measurable sets, then

$$\mu(\bigcup_{n \geq 1} E_n) = \sum_{n=1}^{\infty} \mu(E_n).$$

We call a triple (X, A, μ), with A a σ-algebra and μ a measure, a *measure space*. When appropriate, we omit the triple and simply call X a measure space.

Exercise 8.8. Show that if there is any measurable set E such that $\mu(E)$ is finite, then (M2) follows from (M3).

Exercise 8.9. Suppose E_1, E_2 are measurable and $E_1 \subset E_2$. Show that $\mu(E_1) \leq \mu(E_2)$. (You will almost certainly prove something independently interesting along the way.) Is equality possible?

It is very convenient for us to allow the possibility of sets with infinite measure, not only because they come up in nature but also because it prevents us from stressing about convergence of the sum in (M3). If we drop the requirement (M1), we call μ a *signed measure*.

Definition 8.10. Let X be a measure space and let P be a property that a point $x \in X$ may or may not have. If P holds for all x outside of a set of measure 0, we say that P holds *almost everywhere*, which we frequently abbreviate *a.e.* For instance, "$f(x) = 0$ a.e." means that $f(x) = 0$ for all x except for those contained in a set of measure 0, i.e. $\mu(\{x : f(x) \neq 0\}) = 0$. (Thus "$P$ holds a.e." means $\mu(\{x : \neg P(x)\}) = 0$.)

At last, we can formalize the discussion about measures on $\mathbb{R}$ from the beginning of the section.

Proposition 8.11. *Give $\mathbb{R}$ the Borel σ-algebra. Then there exists a unique measure μ on $\mathbb{R}$ such that $\mu((a, b)) = b - a$, called the* Lebesgue measure.

Exercise 8.12 (for the skeptics)**.** Find and read a proof of the existence of Lebesgue measure.

Exercise 8.13. Let $E \subset \mathbb{R}$ be countable. Show that E is measurable (with the Borel σ-algebra on $\mathbb{R}$) and that the Lebesgue measure of E is 0.

Exercise 8.14. Let $\emptyset \neq U \subset \mathbb{R}$ be open. Show that the Lebesgue measure of U is nonzero. From this, deduce that if $f : \mathbb{R} \to \mathbb{R}$ is 0 almost everywhere, and f is continuous, then in fact f is identically 0. (Hint: $\{x : f(x) \neq 0\} = f^{-1}(\mathbb{R}\backslash\{0\})$.)

The property "$x = 0$" is a *closed* property, in the sense that if it fails for x, then it fails for a whole neighborhood around x. Equivalently, the set of points for which it holds is a closed set – the singleton $\{0\}$. Your argument for the preceding exercise can be ported over to show that continuous functions which have a closed property a.e. actually have that property everywhere. Simply replace "$f(x) \neq 0$" with "$f(x)$ does not have property P."

Remark 8.15 (Unimportant aside for algebraic geometers)**.** As remarked earlier (7.25), Hausdorffness is atypical in algebraic geometry, and a symptom of this is that points tend not to be closed. Even more shocking is the phenomenon of *generic points*: a point $\eta \in X$ is generic if its closure is the whole space, i.e. $\overline{\{\eta\}} = X$. In reasonable cases ("irreducible varieties"), we expect to find a generic point.

Suppose a space X has a generic point η. Then a closed property which holds for η holds for the whole space, since it holds on a closed set containing η. This justifies the terminology "generic point" – in nice situations, η carries the properties of a typical point of X.

Remark 8.16 (important)**.** Implicit in our demand for $\mu((a, b)) = b - a$ is the demand that μ be *translation-invariant*, i.e. that $\mu(E) = \mu(E+t)$ for any $t \in \mathbb{R}$, where

$$E + t = \{e + t : e \in E\}.$$

Indeed, we could have demanded that $\mu((0, 1)) = 1$ and μ be translation-invariant and arrived at the same proposition.

The payoff is the **Lebesgue integral**: a machine that turns a notion of measure on X into a notion of integration on X. Here is a description of that machine.

Step 1. Let (X, A, μ) be a measure space. Suppose $f : X \to \mathbb{R}$ is measurable and

$f(x) \geq 0$ for all $x \in X$. Then, define

$$\int_X f(x)\, d\mu(x) := \lim_{n\to\infty} \sum_{k=0}^{\infty} k2^{-n} \cdot \mu(\{x : k2^{-n} \leq f(x) < (k+1)2^{-n}\}).$$

It is often convenient to agree that $f(x)$ may *take* the value ∞ (say, if f is itself some integral whose finiteness we are concerned about), and if it does so on a set of positive measure, we declare the integral to be ∞ as well.

The right-hand side is called a dyadic sum, owing to the powers of 2 involved. Since $f(x) \geq 0$, $\int_X f(x)\, d\mu(x)$ always exists, if we allow the possibility that it is ∞. Measurability of f ensures that the sets in the summation are measurable. (Check this if you are unsure.) When no ambiguity is possible, we write $\int_X f\, d\mu$ for $\int_X f(x)\, d\mu(x)$.

Step 2. Suppose $f : X \to \mathbb{R}$ is measurable, and set $f^+(x) = \max\{f(x), 0\}$ and $f^-(x) = \max\{-f(x), 0\}$. That is, f^+ is the nonnegative part of f, and f^- is the nonpositive part. Observe that $f(x) = f^+(x) - f^-(x)$ for all x, and that f^+ and f^- are both nonnegative and measurable (why?).

Now, define

$$\int_X f(x)\, d\mu(x) = \int_X f^+(x)\, d\mu(x) - \int_X f^-(x)\, d\mu(x).$$

As long as not both of the integrals on the right-hand side are ∞, this makes sense (allowing for $\infty + a = \infty, a - \infty = -\infty$). If both of $\int_X f^+(x) d\mu(x)$ and $\int_X f^-(x)\, d\mu(x)$ are finite, we call f **integrable**.

Step 3. If $f : X \to \mathbb{C}$ is measurable (with respect to the Borel σ-algebra on $\mathbb{C}$) and $\mathrm{Re} f$ and $\mathrm{Im} f$ are integrable, we define

$$\int_X f(x)\, d\mu(x) = \int_X \mathrm{Re} f(x)\, d\mu(x) + i \int_X \mathrm{Im} f(x)\, d\mu(x). \tag{8}$$

Definition 8.17. Let $E \subset X$ and write 1_E for the indicator function of E, i.e. $1_E(x) = 1$ if $x \in E$ and 0 if $x \notin E$. Then we define

$$\int_E f(x)\, d\mu(x) = \int_X 1_E(x) f(x)\, d\mu(x). \tag{9}$$

Exercise 8.18 (surprisingly tricky but surprisingly useful)**.** Show that the integral of an indicator function 1_E is the measure of E.

A useful consequence is the following: call a function f *simple* if $f = \sum_{i=1}^{n} a_i 1_{E_i}$ is a linear combination of indicator functions. Since integration is linear,

$$\int_G f \, d\mu = \sum_{i=1}^{n} a_i \mu(E_i).$$

We could have defined the integral of any f as the limit of approximations of f by simple functions (e.g. for nonnegative functions, take the infimum of integrals of simple functions s with $s(x) \geq f(x)$ a.e.). Then the preceding exercise would be part of the definition, but certain other arguments would become trickier.

Exercise 8.19 (unexpectedly easy)**.** Check that over the interval $[0, 1]$, $1_{\mathbb{Q}}$ is not Riemann integrable but it is Lebesgue integrable.

Exercise 8.20 (easy)**.** Let $f : X \to Y$ be a continuous function between topological spaces, and give X and Y their Borel σ-algebras. Show that f is measurable.

This machine is sufficiently abstract (requiring only a measure on X) to allow us to define integration over locally compact abelian groups. The problem is, what measure to use? It should understand both the group structure and topology (which itself sees the group structure too). The crucial fact is that a handful of mild demands in this vein leave us with only one measure to use.

Definition 8.21. Let G be a locally compact abelian topological group, and give G the Borel σ-algebra. Suppose μ is a measure on G satisfying the following axioms.

(H1) μ is translation-invariant, i.e. $\mu(E + g) = \mu(E)$ for any measurable E and $g \in G$.

(H2) If $K \subset G$ is compact, then $\mu(K) < \infty$. (In particular, K is measurable, since it is closed in the Hausdorff space G.)

(H3) $\mu(E) = \inf\{\mu(U) : E \subset U, U \text{ open}\}$. This is called **outer-regularity**.

(H4) $\mu(E) = \sup\{\mu(K) : K \subset E, K \text{ compact}\}$. This is called **inner-regularity**.

Then we call μ a **Haar measure** on G.

Axiom (H1) forces μ to "see the group structure," while (H2)-(H4) are concerned with the topology. Though this may seem lopsided, keep in mind that the topology is informed by the group structure, as multiplication and inversion must be continuous.

Theorem 8.22 (as promised)**.** *Let G be a locally compact abelian group. Then there exists a Haar measure on G. Furthermore, let μ and λ be two Haar measures on G. Then $\mu = c \cdot \lambda$ for some $c \in \mathbb{R}$, i.e. $\mu(E) = c\lambda(E)$ for every measurable subset E. That is, Haar measure is unique up to scaling.*

Some examples will illustrate that the Haar measure is incredibly natural. Before reading the following examples, think about what the Haar measure should be on the groups we have worked with so far: $\mathbb{R}, S^1, \mathbb{Z}, \mathbb{Z}_N$.

Example 8.23. On $\mathbb{R}$, Haar measure is Lebesgue measure. Likewise, on S^1, Haar measure is Lebesgue measure when thinking of $S^1 = \mathbb{R}/\mathbb{Z} \cong [0,1) \subset \mathbb{R}$ with the appropriate topology. If we took $S^1 = \mathbb{R}/\gamma\mathbb{Z} \cong [0,\gamma)$ instead, for some $\gamma \in \mathbb{R}$, we would scale the Haar measure by γ.

Example 8.24. On $\mathbb{Z}$ and $\mathbb{Z}_n$, Haar measure is the so-called *counting measure*: $\mu(E) = |E|$. In particular, every subset is measurable, and integration is summation over the elements of the group. Observe that $\mu(\mathbb{Z}_n) = n < \infty$ since $\mathbb{Z}_n$ is compact, but $\mu(\mathbb{Z}) = \infty$ since $\mathbb{Z}$ is not compact (c.f. Exercise 7.47).

Remark 8.25 (promised in Remark 6.7)**.** For compact groups G, the total Haar measure $\mu(G)$ is finite by requirement (H2), and so we can scale μ so as to set $\mu(G) = 1$. This is a fine convention for many purposes (like on the circle), but we will not use it for finite abelian groups. There, the counting measure is so natural that it would be unsettling to carry around a factor of $\frac{1}{|G|}$ in front of *all* of our sums. Keeping track of this factor elsewhere is the price we pay.

(If we took $\mu(G) = 1$, we would have $\int_G f\, d\mu = \frac{1}{|G|}\sum_{g\in G} f(g)$. Then the characters would be genuinely orthonormal, and the Fourier coefficients would be $\hat{f}(e) = \langle f, e\rangle$.)

Exercise 8.26 (mandatory)**.** Let $f : G \to \mathbb{R}$ be a measurable function. Show that, if $f(x) \geq 0$ for all x and $\int_G f\, d\mu = 0$, then f is 0 almost everywhere. Deduce that, if $\int_G |f-g|^p\, d\mu = 0$ for any natural number $p > 0$, then $f = g$ a.e. (Hint for the first part: write $\{x : f(x) > 0\} = \bigcup_{n>0}\{x : \frac{1}{n-1} \geq f(x) > \frac{1}{n}\}$.)

The previous exercise justifies a change in perspective: we should only think of functions as defined almost everywhere, since we can't detect differences on sets of measure 0. For functions on finite abelian groups, this is no change at all, since Haar measure is counting measure so nonempty sets have positive measure. But for groups like S^1 or $\mathbb{R}$, this is a useful new idea.

Definition 8.27. Let G be a locally compact abelian group with Haar measure μ. The vector space $L^p(G;\mathbb{C})$ is the set of measurable functions $f : G \to \mathbb{C}$ such that $\int_G |f|^p\, d\mu < \infty$, modulo the relation $f \sim g$ if $f = g$ almost everywhere, with the usual notion of addition and $\mathbb{C}$-scaling. One can also define $L^p(G;\mathbb{R})$ as a real vector space in the same way. We will rarely feel the need to draw a distinction.

Exercise 8.28 (for the neurotic reader)**.** Check that $L^p(G)$ has well-defined operations, i.e. that if $f_1 \sim f_2$ and $g_1 \sim g_2$, that $\alpha f_1 + f_2 \sim \alpha g_1 + g_2$ for $\alpha \in \mathbb{C}$.

Example 8.29. As we have remarked, this gives us the same notion of $L^2(G)$ for finite abelian groups G that we are already comfortable with. Indeed, inspired by that case, we define an inner product on $L^2(G)$ by

$$\langle f, g \rangle = \int_G f(x)\overline{g(x)}\, d\mu(x). \tag{10}$$

Exercise 8.30. Check that the L^2 condition $\int_G |f|^2\, d\mu < \infty$ ensures that the inner product above is well-defined, i.e. the integral always converges. (Hint: Cauchy-Schwarz.)

Exercise 8.31 (more neuroticism)**.** Check that the inner product of two functions in L^2 does not depend on choice of representatives for them. While you're at it, check that this defines a genuine inner product on $L^2(G)$. The crucial point is that $||f||^2 = \int_G |f(x)|^2\, d\mu(x) = 0$ if and only if $f = 0$ a.e. (c.f. Exercise 8.26).

Now, the long-promised facts:

Theorem 8.32 (Fourier Inversion and Plancherel)**.** *Let G be a locally compact abelian group with Haar measure μ. Let $f \in L^2(G)$ be continuous with compact support (that is, the closure of $\{x \in G : f(x) \neq 0\}$ is compact). Then, with*

$$\hat{f}(e) = \int_G f(x)\overline{e(x)}\, d\mu(x),$$

we have $\hat{f} \in L^2(\widehat{G})$, and there exists a unique Haar measure ν (i.e. a choice of normalization of Haar measure) on $\widehat{G}$ such that

$$f(x) = \int_{\widehat{G}} \hat{f}(e)e(x)\, d\nu(e)$$

almost everywhere. Furthermore, the set of continuous compactly-supported functions is dense in $L^2(G)$, and so the assignment $f \mapsto \hat{f}$ extends to a unique continuous linear map

$$\mathcal{F} : L^2(G) \to L^2(\widehat{G})$$

and $\mathcal{F}$ is an isometry.

Remark 8.33. The measure ν in the Fourier inversion theorem is called the dual measure of μ, and is usually written $\widehat{\mu}$. For instance, when we take counting measure on $\mathbb{Z}_n$ with $\mu(\mathbb{Z}_n) = n$, we need $\widehat{\mu}$ to be $\frac{1}{n}$ times counting measure

for the constants to work out. If we take $\mu(\mathbb{Z}_n) = 1$, then $\widehat{\mu}(\widehat{\mathbb{Z}_n}) = n$, i.e. $\widehat{\mu}$ is counting measure instead. Compare with Remark 8.25. The "self-dual" choice of Haar measure on a finite abelian group G is $\frac{1}{\sqrt{|G|}}$ times counting measure – check this if you like.

Theorem 8.34 (Duality)**.** *Let G be a locally compact abelian group. Then* $\mathrm{ev} : G \to \widehat{\widehat{G}}$ *is a natural isomorphism of groups, and a homeomorphism of topological spaces.*

The proofs of these theorems are much more difficult in the generality that we have now stated them. But the facts are identical to those for finite abelian groups, which should ease your mind.

Exercise 8.35. Let G be a locally compact abelian group with Haar measure μ and let $f \in L^2(G)$. Give $\widehat{G}$ the dual measure $\widehat{\mu}$ as in the Fourier inversion theorem 8.32. Show that $\hat{\hat{f}}(x) = f(-x)$. More properly, show that

$$\hat{\hat{f}}(\mathrm{ev}_x) = f(-x).$$

(Hint: apply the Fourier inversion theorem to $f(-x)$.)

Deduce that applying the Fourier transform four times yields the identity, up to the natural isomorphism $G \cong \widehat{\widehat{G}}$. (You might want to go one step further: when $G \cong \widehat{G}$, we can view the Fourier transform as a linear operator $\mathcal{F} : L^2(G) \to L^2(G)$. What are its possible eigenvalues?)

We are now ready to introduce a new player, which differs from L^2 for cases that we only have the tools to discuss now.

Example 8.36. The vector space $L^1(G)$ was already defined above. But we can define a new operation on it, the *convolution product*, by

$$(f * g)(y) = \int_G f(x)g(y-x)\, d\mu(x). \tag{11}$$

Exercise 8.37 (even more neuroticism)**.** Check that the L^1 condition guarantees that the integral (11) converges for almost all y. (Hint: you will probably need Theorem 8.42. Feel free to invoke it.)

Exercise 8.38 (important details)**.** Check that the vector space $L^1(G)$ becomes an algebra with the product given by convolution.

Exercise 8.39 (reality check)**.** Show that $L^2(S^1) \subset L^1(S^1)$ as vector spaces, but that the converse inclusion does not hold. Show that neither inclusion holds for $L^1(\mathbb{R})$ and $L^2(\mathbb{R})$.

(In fact, the situation is much worse than you might imagine: $L^2(S^1) \subset L^1(S^1)$ as sets, so as vector spaces, but *not* as normed spaces with their respective norms. Their norms are not even *equivalent* when restricted to the subspace $L^2(S^1)$, and L^2 is not a closed subspace of L^1.)

Exercise 8.40 (important)**.** Define the *group algebra* $\mathbb{R}[G]$ as the set of formal (finite) linear combinations

$$\sum_{i=1}^{n} a_i g_i \tag{12}$$

for $a_i \in \mathbb{R}, g_i \in G$. The analytically inclined reader might consider all formal linear combinations

$$\sum_{g \in G} a_g g, \tag{13}$$

possibly with infinitely many nonzero coefficients, but with $\int_G |a_g| \, d\mu(g) < \infty$. We make $\mathbb{R}[G]$ first into a vector space by the expected scaling and addition, namely

$$\alpha \cdot (\sum_{i=1}^{n} a_i g_i) = \sum_{i=1}^{n} (\alpha \cdot a_i) g_i \tag{14}$$

and

$$\sum_i a_i g_i + \sum_i b_i g_i = \sum_{i=1} (a_i + b_i) g_i. \tag{15}$$

Lastly, give $\mathbb{R}[G]$ a product by distributivity:

$$(\sum_i a_i g_i)(\sum_j b_j g_j) = \sum_{i,j} a_i b_j g_i g_j \tag{16}$$

where the product $a_i b_j$ is interpreted in $\mathbb{R}$ and the product $g_i g_j$ is interpreted in G. For example,

$$(a_1 g_1 + a_2 g_2) \cdot (b_1 g_1' + b_2 g_2') = a_1 b_1 g_1 g_1' + a_1 b_2 g_1 g_2' + a_2 b_1 g_2 g_1' + a_2 b_2 g_2 g_2'$$

which should remind you of FOIL. If it concerns you, first check that this structure makes $\mathbb{R}[G]$ into an algebra. Then, show that, for a finite abelian group G, $L^1(G; \mathbb{R}) \cong \mathbb{R}[G]$ as algebras.

(Do not be misled by the finiteness hypothesis; it is to avoid worrying about convergence of the sums (16) defining the coefficients of the product in $\mathbb{R}[G]$.)

Remark 8.41. In particular, if G is a finite cyclic group generated by $x \in G$, then an element of $\mathbb{R}[G]$ can be written as a polynomial in x, with multiplication understood appropriately. This perspective will be relevant when we read [GM95].

Here is a useful fact which should be unsurprising to the reader with some experience in multivariable calculus. We will state it without proof so that you can use it freely when computing integrals later.

Theorem 8.42 (Fubini)**.** *Let $f : X \times Y \to \mathbb{R}$ be measurable and suppose X, Y are σ-finite. Then*

$$\int_{X\times Y} f(x,y)d\mu(x,y) = \int_X \left(\int_Y f(x,y)\, d\mu(y)\right) d\mu(x) = \int_Y \left(\int_X f(x,y)\, d\mu(x)\right) d\mu(y)$$

provided any one of the three integrals are finite. If f takes only nonnegative values, even this last provision is unnecessary – if any of the three integrals are ∞, so are the others.

The technical condition of σ-finiteness should be of no concern. All the groups we have seen or will work with are σ-finite.

Exercise 8.43 (for the untrusting reader)**.** Look up the definition of a σ-finite measure space and check that the following groups are σ-finite: $\mathbb{R}^n, S^1, \mathbb{Z}, \mathbb{Z}_n$. In fact, more is true: every connected locally compact abelian group is σ-finite.

One more word of caution: to be totally precise, we should specify a *product measure* on $X \times Y$, given measures on X and Y. But we will only be concerned with locally compact abelian groups X, Y with their Haar measures. In such a case, $X \times Y$ is also a locally compact abelian group, and the induced measure is, as you might have guessed, the Haar measure on $X \times Y$.

As an application, here is a fun and useful exercise.

Exercise 8.44. Let $f, g \in L^1(G)$. Prove the **convolution theorem**:

$$\widehat{f * g} = \hat{f} \cdot \hat{g}.$$

Where did you use Fubini's theorem?

This has many practical uses. For one, we have a remarkably fast algorithm for computing the Fourier transform of a function on $\mathbb{Z}_n$ (called, aptly enough, the Fast Fourier Transform). As it happens, it is more efficient (for n large) to compute $\hat{f}$ and $\hat{g}$, multiply them pointwise, and invert the Fourier transform,

rather than to compute the convolution directly.

On the more theoretical side, the convolution theorem expresses that the Fourier transform is a homomorphism of algebras between $L^1(G)$ and... the algebra $\mathcal{A}$ of functions which are the Fourier transforms of something in $L^1(G)$. No satisfactory characterization of $\mathcal{A}$ is known.

Note that we could have *defined* convolution to be the thing which corresponds to multiplication on the Fourier side. That is, we could have defined

$$f * g = \mathcal{F}^{-1}(\mathcal{F}(f) \cdot \mathcal{F}(g))$$

where, as before, $\mathcal{F}$ is the linear operator $f \mapsto \hat{f}$. It would then be a computation to check that this is the convolution we are familiar with. In the same vein, we might define

$$f \star g = \mathcal{F}^{-1}(\mathcal{F}(f) * \mathcal{F}(g))$$

so that $\star$ is the operation which is sent to convolution by the Fourier transform. It turns out that you are already familiar with $\star$. What is it?

9 Grünbaum-Moore

The following exercises are a companion to the paper [GM95]. Consult them regularly as you read, and think about them as they come up.

Remark 9.1. Grunbaum and Moore use the following slightly unusual conventions. First of all, they think of the finite cyclic group of order n as μ_n, the group of roots of unity, rather than $\mathbb{Z}_n$. Secondly, for them, a function $f : \mathbb{Z}_n \to \mathbb{R}$ is a "sequence" (i.e. vector) $(c_1, ..., c_n)$ where $c_i = f(i)$. That is, they identify $L^2(\mathbb{Z}_n)$ with $\mathbb{R}^n$ by taking the standard basis $\{f_i(j) = \delta_{ij}\}$ of L^2. Then they write $\hat{f}(i)$ as d_i, so the sequence (d_i) is the Fourier transform of the sequence (c_i).

There are a great many different perspectives on autocorrelations, and they each come with their own notation. The exercises below are meant to unify the notation where possible, and to guide you to making sense of it otherwise.

Exercise 9.2 (mandatory)**.** On p. 313, Grünbaum and Moore assert that two expressions "are related by a multiple Fourier transform." Check this. More specifically, check that, for

$$\rho_n(f)(t_1, ..., t_{n-1}) = \sum_{x \in \mathbb{Z}_N} f(x)f(x + t_1)...f(x + t_{n-1}),$$

we have

$$\widehat{\rho_n(f)}(e_1, ..., e_{n-1}) = \hat{f}(e_1)\hat{f}(e_2)...\hat{f}(e_{n-1})\hat{f}(e_1^{-1}...e_{n-1}^{-1}).$$

Your argument will extend to any locally compact abelian group without change (except for reinterpreting the sum as an integral with respect to Haar measure). Note also that we are choosing counting measure on $\mathbb{Z}_N$ – but you should argue without worrying about normalization factors.

Exercise 9.3 (mandatory)**.** Check that the so-called "translation invariants" are indeed translation-invariant. That is, if $g(x) = f(x+a)$ for some fixed a and every x, check that f and g have the same translation invariants. (Hint: don't check this on the Fourier side. Check it using the definition of the translation invariants as a sum as in Exercise 9.2.)

Exercise 9.4 (mandatory)**.** Check the effect of translation on the Fourier side: if $f, g : \mathbb{Z}_N \to \mathbb{R}$ are functions such that $f(x) = g(x+a)$ for all x (for some fixed $a \in \mathbb{Z}_N$), what is the relationship between $\hat{f}$ and $\hat{g}$? Use this to solve Exercise 9.3 in a second way.

Exercise 9.5 (optional)**.** Let 1_E be the indicator function of some subset $E \subset G$. Define the *difference multiset*

$$E - E = \{(e, m) : e \text{ appears } m \text{ times as a difference } e_1 - e_2 \text{ for } e_1, e_2 \in E\}.$$

Check that $\rho_2(1_E)$ carries the same data as $E - E$. In particular, if the multiplicity of e in $E - E$ is m, check that $\rho_2(1_E)(e) = m$. Use this to make sense of the examples Grunbaum and Moore present on p. 314 as part of the story of Bloom and Piccard.

The story of Piccard's theorem and Bloom's counterexample is an interesting one. Piccard believed that she proved that a set was determined by its difference multiset, under certain conditions [Pic39]. Bloom presented a counterexample on a "6-mark ruler," i.e. on $\mathbb{Z}_6$ [Blo77]. It was later shown by Bekir and Golomb that there are no other counterexamples [BG07].

Exercise 9.6 (optional)**.** Do Exercise 8.40, read the remark immediately following it, and make sense of the examples of polynomials that Grunbaum and Moore present on p. 313-314.

Exercise 9.7. At the beginning of Section 5.1, Grünbaum and Moore cite the Chinese Remainder Theorem to prove the following fact with no further argument. Prove it in detail, with or without the Chinese Remainder Theorem. (Hint: consider cyclotomic polynomials instead. The Fourier transform of a function on $\mathbb{Z}_N$ can be viewed as a polynomial in a primitive root of unity.)

Proposition 9.8. *Let $f : \mathbb{Z}_N \to \mathbb{Q}$ be a function. If $\hat{f}(1) \neq 0$, then $\hat{f}(k) \neq 0$ for any k relatively prime to N.*

Exercise 9.9. On p. 315, Grunbaum and Moore call the following fact "just a consequence of the result about 'multiplication of residue classes'". Prove it in detail. (Hint: prove it for $N = p^k$ a prime power, and then apply the Chinese Remainder Theorem. You may find it useful to split the cases $p = 2$ and $p > 2$.)

Proposition 9.10. *Let N be an integer. Any even integer can be written as the sum of two integers, each relatively prime to N. In particular, $\mathbb{Z}_N^\times + \mathbb{Z}_N^\times = 2\mathbb{Z}_N$.*

Deduce that every element of $\mathbb{Z}_N$ is a sum of at most 3 elements of $\mathbb{Z}_N^\times$ for **any** N.

Exercise 9.11. On p. 311, Grunbaum and Moore write: "Of course, equality of first- and sixth-order invariants implies equality for all orders in between." Give an example to show that equality of first- and third-order invariants does *not* imply equality of second-order invariants. Prove that, when $\hat{f}(0) \neq 0$, knowledge of first- and nth-order invariants lets you compute the intermediate invariants explicitly. Is the claim of Grunbaum and Moore true when $\hat{f}(0) = 0$? (This last question I do not know the answer to.)

Exercise 9.12. Let $N \geq 6$. Given a function $f : \mathbb{Z}_N \to \mathbb{C}$, describe an algorithm to reconstruct f from its Nth order invariants.

Exercise 9.13 (follow-up)**.** Give an example of a pair of functions $f, g : \mathbb{Z}_N \to \mathbb{C}$ which are not translations of each other but share autocorrelation data through order $N - 1$. Are they both $\mathbb{R}$-valued?

10 Adler-Konheim

Exercise 10.1. Suppose $f \in L^1(\mathbb{R})$. Show that $\hat{f}$ is continuous. (Hint: Sequential continuity. At some point, you will want to interchange a limit and an integral; to justify this, you will need the so-called Dominated Convergence Theorem, or something equivalent, which you should feel free to look up.)

Exercise 10.2 (enlightening)**.** Prove the following lemma.

Lemma 10.3. *Let $f, g \in L^1(G)$ for G a locally compact abelian group. If f and g have the same invariants of order 2 and k, then*

$$\prod_{i=1}^{m} \frac{\hat{f}(x_i)}{\hat{g}(x_i)} = \prod_{j=1}^{n} \frac{\hat{f}(y_j)}{\hat{g}(y_j)} \tag{17}$$

for any $x_i, y_j \in S(f)$ such that $\sum x_i = \sum y_j$ and $m + n = k$.

Exercise 10.4. Prove the corollary on p. 428. (Warning: in our notation, Adler and Konheim mean $3v$ when they write $3v - 1$.)

Hint: it might be illustrative to consider first the case where $\hat{f}$ never vanishes, i.e. $S(f) = \widehat{G}$. For the general case, you will want to use Lemma 10.3; see the proof of the main theorem of Grunbaum-Moore, p. 320-321 of their paper, for the technique. Once you know that $\frac{\hat{f}}{\hat{g}}$ is a character, conclude the proof by extracting the argument from the main theorem of Adler-Konheim. p. 427.

Exercise 10.5. Use the corollary on p. 428 (which you proved in Exercise 10.4) to prove that a function $f : \mathbb{Z}_N \to \mathbb{Q}$ is determined by its autocorrelation data through order 9. Is this bound sharp? (Hint for the first part: Exercises 9.7 and 9.9.)

By considering the cases N even and N odd separately, refine the result of this exercise. Is that bound sharp?

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